Business & Domain Organization Strategy

Ami Heines — Blockchain Infrastructure Consulting

March 2026

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Business & Domain Organization Strategy for Ami Heines

Context

Ami has multiple domains (amiheines.com, pofov.com, chartly.com, verarta.com), several deployed apps (ChainVault, Verarta, PizzaDiv), and self-hosted infrastructure (MiaB, Listmonk, Matomo). He wants to organize everything into a coherent presence with one main hub, reach $2K/month revenue as fast as possible, and build toward $5-15K/month through a mix of direct outreach and content-driven inbound. He has 20+ hours/week and is comfortable on camera. He uses Claude Code as an AI-accelerated development partner to build, expand, and monitor all projects — and demonstrates this workflow to prospects as proof of delivery speed.

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1. Domain Strategy

amiheines.com → Main Hub - Personal name domains are the strongest asset for solo consulting. Customers buy from a person. - When someone watches a YouTube video and Googles "Ami Heines," this is where they land. - MiaB stays at mail.amiheines.com, Listmonk at lists.amiheines.com — no conflict.

Other domains:

Domain	Action
pofov.com	**SEO lead-gen engine** — crypto/blockchain research site with 1000+ coin pages. Captures long-tail search traffic, gates high-value content behind email, funnels blockchain-curious visitors to amiheines.com consulting. Also usable as YouTube brand ("POFOV — Proof of Value").
chartly.com	Park or sell — no connection to consulting positioning
verarta.com	Keep as its own project, link from portfolio
privateblockchaindatabase.com	301 redirect to amiheines.com/blog, migrate the 16 existing posts

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2. Target Customer

Who: CTOs, VPs of Engineering, IT Directors at mid-size companies (50-500 employees) needing tamper-proof records, audit trails, or fraud prevention — without in-house blockchain expertise.

Best verticals: 1. **Real estate title companies** — wire fraud is a $446M/year problem; ChainVault is built for this 2. **Legal firms** — M&A, closings, IP transactions need tamper-proof document trails 3. **Art/provenance**

tracking — Verarta proves the capability 4. **Healthcare compliance** — HIPAA audit trail requirements 5. **Government/municipal IT** — transparency and public records

Key positioning: Don't sell "blockchain consulting." Sell **tamper-proof record-keeping** and **fraud prevention infrastructure**. The blockchain is the *how*, not the *what*.

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3. Main Website (amiheines.com)

Tech: Astro 5 + Tailwind v4 + MDX (reuse the Dante theme from privateblockchaindatabase.com)

Pages:

```
/                   → Landing (hero + value prop + social proof)
/about              → Bio, background, philosophy
/services           → Service packages (all tiers, from $500 assessment to $5K PoC)
/portfolio          → Case studies: ChainVault, Verarta
/blog               → Blog posts (migrated + new, 1/week)
/contact            → Contact form + Cal.com booking embed
/newsletter         → Listmonk signup
/resources          → Gated lead magnets (email required)
```

Deploy: Same server (158.69.27.200), nginx + static Astro output, certbot SSL.

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4. Revenue Strategy — Outbound First, Content Second

The fastest path to $2K/month is **direct outreach**, not waiting for content to generate inbound leads. Content builds the long-term flywheel, but outbound generates revenue now.

Time split (first 3 months): 70% outreach & sales / 30% content **Time split (months 4+):** 50/50, shifting toward content as inbound picks up

Direct Outreach (weeks 1-10)

LinkedIn outreach cadence: 10-15 personalized messages/week to decision-makers in target verticals.

Message framework: 1. Reference something specific about their company (a recent transaction, a compliance concern in their industry, a news article) 2. Name the problem in 1-2 sentences (wire fraud exposure, audit trail gaps, document integrity risk) 3. Offer a free 30-minute audit / discovery call — no pitch, just value 4. Link to amiheines.com for credibility

Target lists to build: - 50 real estate title company CTOs / IT directors - 30 legal firm technology partners / CIOs - 20 healthcare compliance officers - 20 government IT directors

Warm outreach: Post in relevant LinkedIn groups, comment on target prospects' posts for 2 weeks before DMing. Join relevant Slack communities, Discord servers, and forums where these buyers discuss problems.

Freelance Platforms (bridge revenue)

While outbound pipeline builds, list on: - **Toptal** (expert-vetted, high rates) - **Upwork** (expert-vetted tier) - **Catalant** (enterprise consulting marketplace)

Position as: blockchain infrastructure / tamper-proof records specialist. Even 10-15 hrs/month at $100-150/hr = $1,000-2,250/month while the direct pipeline matures.

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5. Service Packaging

Tier 1: $500 — Architecture Review (2 hours)

- Recorded video walkthrough of how tamper-proof records would fit their existing stack

- Written summary with go/no-go recommendation
- **Why it works:** Low commitment, fast close, builds trust for upsell

Tier 2: $997 — Feasibility Assessment (1 week)

- Written report: current fraud/integrity exposure, recommended architecture, cost estimate, implementation roadmap
- 30-minute walkthrough call
- **Why it works:** Concrete deliverable, justifiable expense for any department budget

Tier 3: $1,500 — Half-Day Engineering Workshop

- Live workshop for engineering team: private blockchain fundamentals, architecture patterns, hands-on demo
- Recording + slides delivered after
- **Why it works:** Multiple stakeholders see the value, creates internal champions

Tier 4: $5,000 — Private Chain Proof of Concept (flagship)

Week 1-2: Discovery + architecture document **Week 3-4:** Working PoC (2-node chain, 1 smart contract, basic web UI, API docs) **Deliverables:** Source code (MIT), Docker Compose, architecture doc, demo walkthrough, 30-min follow-up

Why $5K works: CTO can approve without board; filters tire-kickers; leads to $20-50K full implementation.

Tier 5: Ongoing / Upsell

- $500/hour consulting
- $15K blockchain architecture audit
- $20-50K full implementation
- $2-5K/month managed private blockchain infrastructure

Upsell path: $500 review → $997 assessment → $5K PoC → $20-50K implementation → $2-5K/month managed infra

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6. Content Strategy

YouTube (1 video/week — not 2, to protect outreach time): - **"Build With Me"** (15-30 min) — deploying private chains, building encrypted vaults, real infrastructure - **"Explain Like I'm a CTO"** (8-15 min) — wire fraud, audit trails, why your sysadmin shouldn't be able to edit logs - **"Behind the Infrastructure"** (5-10 min) — self-hosted stack tours, production topology

Blog: 1 post/week, derived from video content. SEO-targeted.

LinkedIn: Daily short-form insights (5 min/day). This is where $5K customers actually are. Comment on prospects' posts. Share opinions, not just tutorials.

Funnel:

```
YouTube / LinkedIn / Blog → lead magnet download (email gate) → Listmonk signup
  → weekly newsletter (value + soft pitch)
  → every 4th email: "Book a free 30-min discovery call"
  → discovery call → proposal (right tier for their stage)
```

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7. Mailing List (Listmonk)

Lead magnets: - "Private Blockchain Infrastructure Checklist" - "Wire Fraud Prevention Playbook for Real Estate" - "Audit Trail Compliance Guide"

Welcome sequence (5 emails over 14 days): 1. Welcome + best resource 2. Your story + why private blockchain 3. Case study (ChainVault) 4. Common misconceptions 5. “How I can help” + book a call CTA

Ongoing: Weekly newsletter + monthly case study deep dive

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8. Timeline to $2K/month

Week	Action	Expected Results
1-2	Website live, LinkedIn profile optimized, 1 lead magnet created, freelance platform profiles submitted	Foundation in place
2-4	Build prospect lists (120+ contacts). Send 30-40 personalized LinkedIn messages. First 2 blog posts live.	3-5 discovery calls booked
4-6	Close first paid engagements ($500-$1,500 tier). First YouTube video published. Continue outreach (10-15/week).	$500-$1,500 revenue
6-8	Freelance platform revenue starts. Refine pitch based on call feedback. 4+ videos live.	$1,000-$2,000/month
8-12	Referrals from first customers. Repeat engagements. Content pipeline running. Inbound leads begin.	$2,000-$3,000/month
12-16	Upsells to $5K PoC tier. Managed infra contracts. Content flywheel accelerating.	$3,000-$5,000/month
16-24	40+ videos, strong SEO, testimonials, inbound dominant. Shift to 30% outreach / 70% content.	$5,000-$15,000/month

Realistic answer: 6-10 weeks to first revenue, 8-12 weeks to $2K/month, with consistent 20+ hrs/week split between outreach and content.

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9. Implementation Steps

Phase 1: Main Website (2-3 days)

- Fork/adapt the Dante Astro theme from privateblockchaindatabase.com
- Add /services (with all tiers), /portfolio, /contact, /resources pages
- Integrate Listmonk API for newsletter signup
- Embed Cal.com for booking
- Add Matomo tracking
- Migrate 16 blog posts from privateblockchaindatabase.com
- Deploy to server (nginx + certbot)
- Set up 301 redirects from privateblockchaindatabase.com

Phase 2: Lead Magnets (1 day)

- Create first PDF lead magnet
- Build gated /resources page (email → Listmonk API → download)

Phase 3: Listmonk Setup (1 day)

- Create lists, welcome sequence, templates
- Double opt-in configuration

Phase 4: Outreach Infrastructure (2-3 days)

- Optimize LinkedIn profile (headline = problem you solve, not job title)
- Build prospect lists in a spreadsheet/CRM (120+ contacts across 4 verticals)
- Write 3-4 message templates (personalize per prospect)
- Set up Toptal / Upwork / Catalant profiles
- Create a simple discovery call script and proposal template

Phase 5: YouTube + Content Pipeline (ongoing)

- Set up channel, branding, thumbnail template
- Recording workflow: OBS → DaVinci Resolve → YouTube → blog post → LinkedIn → newsletter
- 1 video/week cadence

Phase 6: Analytics (half day)

- Matomo goals: signups, contact submissions, services page visits
- Campaign tracking on YouTube/email links
- Track outreach metrics: messages sent, response rate, calls booked, close rate

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Key Files

- `/home/ami/dev/work/privateblockchaindatabase.com/` — Dante Astro theme to adapt
- `/home/ami/dev/work/privateblockchaindatabase.com/src/content/blog/` — 16 posts to migrate
- `/home/ami/dev/work/chainvault/infra/production/nginx/` — nginx config pattern to replicate
- `/home/ami/dev/work/chainvault/DEPLOYMENT.md` — deployment pattern reference

Verification

- After building: visit amiheines.com, test all pages, newsletter signup flow, contact form
- Verify 301 redirects from privateblockchaindatabase.com
- Test Listmonk API integration (subscribe → receive welcome email)
- Confirm Matomo tracking fires on all pages

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10. Blog Post & Video Topic Ideas

Topics organized by audience and intent. Each can be a blog post, a YouTube video, or both. Prioritize topics marked **[HIGH INTENT]** first — these attract people actively looking to solve a problem (closer to buying).

Wire Fraud & Document Security (Real Estate / Legal)

1. **[HIGH INTENT]** “How Wire Fraud Actually Works — And Why Email Verification Won’t Save You”
2. **[HIGH INTENT]** “Building a Tamper-Proof Closing Document Trail for Title Companies”
3. “The $446M Wire Fraud Problem: What Real Estate Tech Is Missing”
4. “Why Your Document Signing Platform Isn’t Tamper-Proof (And What to Do About It)”
5. “Immutable Audit Trails for M&A Transactions: A Technical Walkthrough”
6. “What Happens When a Notarized Document Gets Altered After Signing?”
7. “Chain of Custody for Legal Documents: Beyond PDF Signatures”

Audit Trails & Compliance (Healthcare / Government)

8. **[HIGH INTENT]** “HIPAA Audit Trail Requirements: What Your Current System Is Probably Missing”
9. **[HIGH INTENT]** “Building Immutable Audit Logs That Survive a Compliance Audit”
10. “Why Your Sysadmin Can Edit Your Audit Logs (And Why That’s a Problem)”
11. “Tamper-Proof Public Records: How Municipal IT Can Restore Citizen Trust”
12. “SOC 2 Audit Trail Requirements: A Practical Implementation Guide”
13. “The Difference Between Append-Only Logs and Truly Immutable Records”

Technical Build-Alongs (Developer / CTO Audience)

14. **[HIGH INTENT]** “Deploying a Private Blockchain in 30 Minutes with Docker Compose”
15. “Building an Encrypted Document Vault with Hyperledger Fabric — Full Walkthrough”
16. “Private Blockchain vs. Traditional Database: When Each One Wins”
17. “Smart Contracts for Document Verification: A Practical Example”
18. “How I Self-Host My Entire Business Infrastructure (Mail, Analytics, Email Marketing)”
19. “Building ChainVault: Architecture Decisions for a Real-World Private Blockchain App”
20. “Zero-Knowledge Proofs for Document Verification — Explained for Backend Engineers”
21. “Private Blockchain Performance: Benchmarking Hyperledger vs. Quorum vs. Corda”
22. “From Docker Compose to Production: Deploying Blockchain Infrastructure on Bare Metal”

Infrastructure & Self-Hosting (Developer Community / Brand Building)

23. “My Full Self-Hosted Stack: MiaB, Listmonk, Matomo, and Why I Ditched SaaS”
24. “Self-Hosting Your Email Server in 2026: Deliverability, DKIM, and Lessons Learned”
25. “Running Matomo Instead of Google Analytics: What I Gained and Lost”
26. “Nginx Reverse Proxy for Multiple Apps on One Server — Production Config Walkthrough”
27. “Listmonk as a Mailchimp Replacement: Setup, API Integration, and Honest Review”
28. “SSL, DNS, and Mail: The Unholy Trinity of Self-Hosted Infrastructure”

Art Provenance & NFT/Tokenization (Niche but Differentiating)

29. “Digital Provenance for Physical Art: How Verarta Works Under the Hood”
30. “Why Art Authentication Needs Blockchain (Not AI)”
31. “Tokenizing Real-World Assets: Beyond the Hype, Here’s What Actually Works”

Business / Strategy (LinkedIn-First Content)

32. “Why I Stopped Saying ‘Blockchain Consultant’ and Started Saying ‘Fraud Prevention Infrastructure’ ”
33. “The $5K Proof of Concept: Why CTOs Love (and Budget For) This Engagement Model”
34. “What Enterprises Get Wrong About Private Blockchains”
35. “Blockchain Is Not Crypto: A Message for Every CTO Who Tuned Out in 2022”
36. “Build vs. Buy for Tamper-Proof Records: A Decision Framework”
37. “Why Mid-Size Companies Are the Sweet Spot for Private Blockchain Adoption”

Case Studies & Stories

38. “How I Built a Wire Fraud Prevention System for Real Estate Closings” (ChainVault deep dive)
39. “From Idea to Production: Deploying a Private Blockchain for Art Provenance” (Verarta story)
40. “The Real Cost of Not Having Immutable Records — 3 Cautionary Tales”

Beginner / Awareness (Top of Funnel — Wide Reach)

41. “What Is a Private Blockchain? (It’s Not What You Think)”
42. “Blockchain Without Crypto: 5 Enterprise Use Cases That Actually Work”
43. “Public vs. Private Blockchain: A 5-Minute Explainer for Non-Technical Leaders”
44. “Do You Actually Need a Blockchain? A Honest Flowchart”
45. “The State of Enterprise Blockchain in 2026: What’s Real, What’s Hype”

SEO Long-Tail (Search-Driven Blog Posts)

46. **[HIGH INTENT]** “best private blockchain platform for enterprise 2026”
47. **[HIGH INTENT]** “how to make audit logs tamper proof”
48. **[HIGH INTENT]** “wire fraud prevention technology real estate”
49. “hyperledger fabric tutorial docker compose”
50. “private blockchain vs permissioned database”
51. “blockchain document verification system”
52. “HIPAA compliant audit trail software”
53. “immutable record keeping for legal documents”

Video Series Concepts

“Explain Like I’m a CTO” (8-15 min, weekly rotation): - Topics 1, 3, 8, 10, 11, 16, 34, 35, 41, 43, 44

“Build With Me” (15-30 min, bi-weekly): - Topics 14, 15, 17, 19, 22, 26, 27

“Behind the Infrastructure” (5-10 min, monthly): - Topics 18, 23, 24, 25, 28

“Case Study Breakdown” (10-20 min, monthly): - Topics 38, 39, 40

POFOV + Blockchain Expertise (Cross-Over Content)

These topics use POFOV’s data, infrastructure, and 1,000+ coin research as a backdrop to demonstrate blockchain expertise. Each piece lives on **both** pofov.com (for SEO reach) and amiheines.com/blog (for consulting credibility), with cross-links. Videos reference POFOV as a live example throughout.

Platform Deep Dives — “I Actually Tested This” Series Each of these uses POFOV’s AI-generated research + your hands-on deployment experience to give an opinion no one else can: you’ve read what 3 AI models say about the platform AND you’ve deployed production systems on it.

54. **[HIGH INTENT]** “Antelope in 2026: What 3 AI Models Say vs What I Found Deploying It in Production” — pull POFOV’s AI analysis of Antelope, then show real metrics from Verarta/ChainVault (block times, finality, throughput). The gap between AI summaries and reality is the content.
55. **[HIGH INTENT]** “Hyperledger Fabric: POFOV Research vs Production Reality — Who Should Actually Use It?”
56. **[HIGH INTENT]** “R3 Corda vs Antelope for Financial Transactions — I Analyzed Both, Here’s What Matters”
57. “ConsenSys Quorum: What Every CTO Should Know Before Committing (POFOV Research + My Take)”
58. “AWS Managed Blockchain vs Self-Hosted Antelope: Cost, Control, and What POFOV’s Data Misses”

Comparison Content — Powered by POFOV Data

59. **[HIGH INTENT]** “I Compared 1,000 Blockchain Projects with AI — Here Are the 7 That Actually Work for Enterprise” — curate POFOV’s dataset into an enterprise shortlist, explain your methodology, link to individual POFOV pages
60. **[HIGH INTENT]** “On-Chain Storage: Which Blockchains Actually Support It? (I Checked All of Them)” — use POFOV AI answers on storage capabilities, verify against docs, show Verarta as proof of on-chain file storage
61. “Smart Contract Support Across 1,000 Chains: What the Data Tells Us” — aggregate POFOV AI answers on programmability, visualize in a chart, discuss what matters for enterprise
62. “Blockchain Transaction Speed Claims vs Reality: What AI Says vs What I Measured” — POFOV AI data on throughput claims, then your production benchmarks from Antelope
63. “The Private Blockchain Landscape in 2026: A Data-Driven Analysis of 50+ Platforms” — POFOV category data + enterprise subset analysis, published as gated PDF on POFOV
64. “Consensus Mechanisms Compared: BFT, PBFT, Raft, Savanna — Which One Should You Choose?” — reference POFOV AI answers for each, then give your production opinion based on running Savanna

Building POFOV — Technical Build-Alongs

65. “How I Built a 1,000-Page SEO Site with Astro, AI, and a Python Scraper” — full architecture walkthrough of POFOV. Demonstrates technical breadth beyond blockchain.

66. “Scraping CoinMarketCap Ethically: Rate Limits, Retries, and Resume Support in Python” — deep dive into the scraper, useful for anyone building data pipelines
67. “Using 3 AI Models (Gemini, Claude, GPT) to Research 1,000 Crypto Projects — Costs, Quality, and Surprises” — compare AI provider quality on the same questions, show real cost data from POFOV’s scrape runs
68. “Building an AI-Powered Comparison Tool for Blockchain Platforms” — how the /compare pages work, from data pipeline to rendering
69. “From 0 to 5,000 Pages: Programmatic SEO with Astro and AI-Generated Content” — the strategy and technical execution behind POFOV’s scale
70. “Adding Email Capture to a 1,000-Page Site Without Ruining the User Experience” — POFOV’s gating strategy, Listmonk integration, conversion rate data

Enterprise Blockchain Evaluation — POFOV as Decision-Making Tool

71. **[HIGH INTENT]** “How to Choose a Private Blockchain Platform: A Framework Using Real Data” — introduce POFOV as a research tool, walk through evaluation criteria (storage, smart contracts, speed, consensus, ecosystem), show how to use POFOV’s comparison and matrix tools
72. **[HIGH INTENT]** “The Enterprise Blockchain Buyer’s Checklist — 20 Questions to Ask Before You Commit” — each question links to POFOV data that helps answer it
73. “Why Most Blockchain Comparisons Are Useless (And How I Built One That Isn’t)” — critique of existing comparison articles (marketing-driven), explain POFOV’s data-driven approach
74. “I Asked AI to Evaluate Every Major Blockchain for Enterprise Use — Here’s the Ranking” — POFOV AI data aggregated into a ranked list with commentary
75. “Blockchain Storage Costs: A Calculator and What the Numbers Actually Mean” — POFOV’s storage calculator tool as the centerpiece, with real cost analysis from running Verarta

Crypto Industry Analysis — POFOV Data as Authority

76. “What 1,000 AI-Analyzed Crypto Projects Tell Us About Where Blockchain Is Actually Going” — trend analysis from POFOV categories: which sectors are growing, which are stagnant
77. “The Rise of RWA (Real World Asset) Tokens: POFOV Data on the 50 Leading Projects” — deep dive into POFOV’s RWA category
78. “Enterprise Blockchain Adoption: 7 Platforms That Companies Are Actually Using (With Evidence)” — POFOV enterprise section + your consulting experience
79. “DeFi, Gaming, AI — Which Blockchain Category Has Real Enterprise Potential?” — POFOV category data analyzed through an enterprise lens
80. “Blockchain Privacy Features Compared: 15 Chains Analyzed for Enterprise Compliance” — pull POFOV AI answers on privacy, overlay with compliance requirements (GDPR, HIPAA)

ChainTalk + POFOV Combined

81. “Why I Added Blockchain Comments to 1,000 Crypto Research Pages (And What Happened)” — results of deploying ChainTalk on POFOV, engagement data, what discussions formed
82. “Building a Blockchain-Powered Discussion Forum for Crypto Research” — ChainTalk on POFOV as case study: moderation challenges, spam handling, community value
83. “Transparent Moderation: What Happens When Every Delete Is a Blockchain Transaction” — real examples from ChainTalk on POFOV, moderation audit trail

LinkedIn Short-Form — POFOV as Proof Point These are 1-2 paragraph LinkedIn posts (5 min to write) that reference POFOV data to establish authority:

84. “I analyzed 1,000 blockchain projects. Only 7 are viable for enterprise. Here’s why…” → link to POFOV
85. “Everyone claims 100K TPS. I checked the actual data on 50 chains. The truth is embarrassing.” → POFOV research
86. “A CTO asked me which blockchain to use. Instead of guessing, I showed them this comparison tool.” → POFOV /compare
87. “3 AI models analyzed the same blockchain. They disagreed on almost everything. Here’s what I learned.” → POFOV
88. “The best blockchain for storing files on-chain? I’ve now tested 3. Here’s my ranking.” → POFOV + Verarta experience

89. “Most blockchain comparison sites are paid placements. I built one that isn’t.” → POFOV
90. “I put immutable comments under 1,000 crypto pages. The community response surprised me.” → ChainTalk + POFOV

Video Series Additions for POFOV “POFOV Research Review” (10-15 min, bi-weekly) — NEW SERIES: Screen-share POFOV, walk through a specific platform or comparison, add your expert commentary. Low production effort, high SEO value. - Topics 54, 55, 56, 57, 58, 59, 60, 62, 64, 74

“Build With Me” additions: - Topics 65, 66, 67, 68, 69, 70

“Explain Like I’m a CTO” additions: - Topics 71, 72, 73, 78, 80

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11. Software Development — Portfolio Expansion with Claude Code

The best way to win consulting customers is to **show working software, not slide decks**. This section covers concrete development work across three projects — Verarta, ChainVault, and a new Parking Tickets system — plus a shared observability layer using 100% open-source tools. Each project doubles as a live demo, a portfolio piece, a blog/video topic, and proof that you deliver fast with AI-assisted development.

Development Philosophy

- **Claude Code as force multiplier.** Every project is built with Claude Code (Claude CLI). This is itself a selling point: “I ship production blockchain systems in weeks, not months, by combining deep domain expertise with AI-accelerated development.”
- **Open-source monitoring stack.** Every project gets a Grafana dashboard backed by Prometheus, Loki, and cAdvisor. Prospects see that enterprise-grade observability doesn’t require $50K/year Datadog contracts.
- **One blockchain cluster, many apps.** Verarta, Parking Tickets, and future clones all run on the same Antelope/Spring 4-node cluster. ChainVault runs its own. This demonstrates both shared-infra and dedicated-infra deployment models.
- **Everything self-hosted.** The full stack runs on a single server (158.69.27.200) — blockchain, apps, monitoring, email, analytics. This is the pitch: “I can give you this entire stack.”

11.1 Shared Monitoring Stack (All Projects)

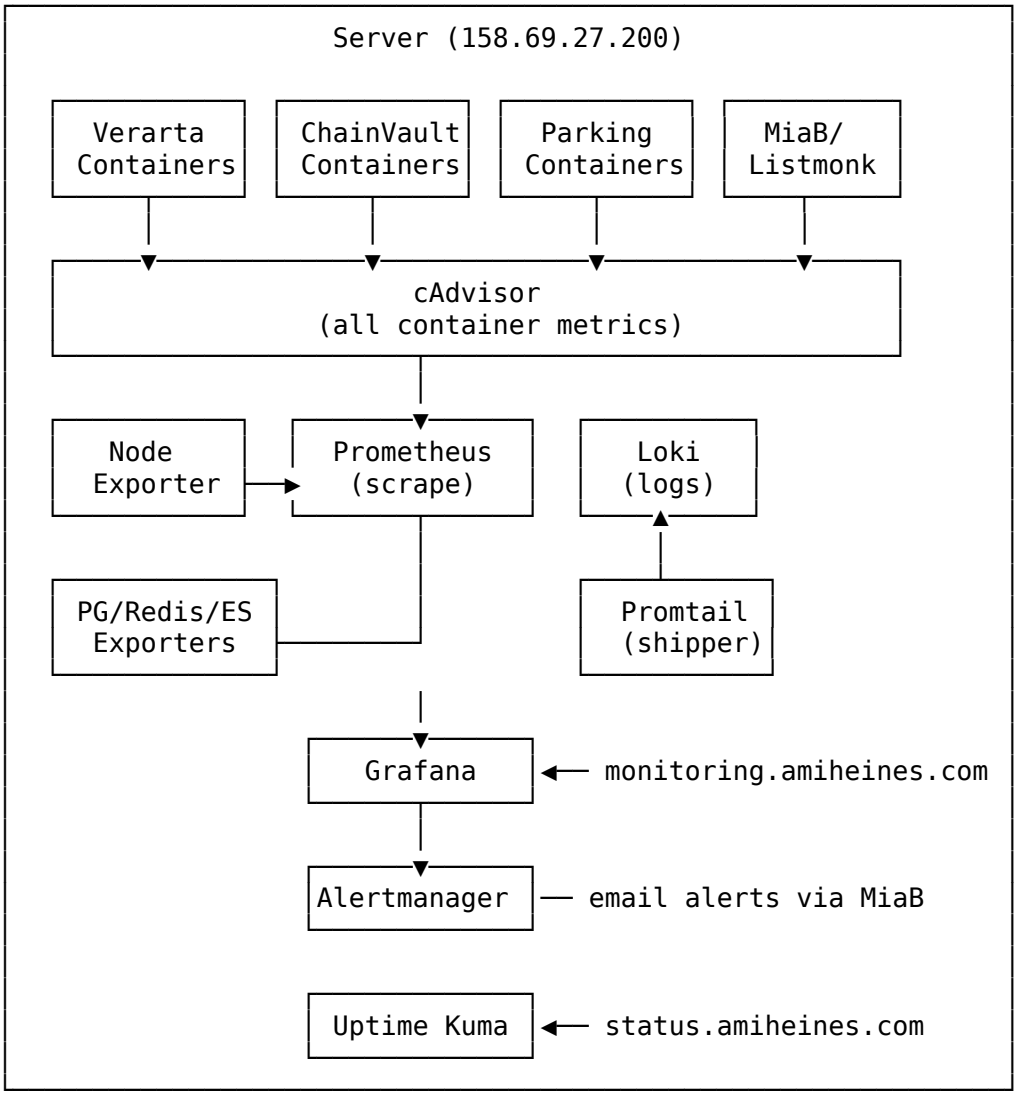
Deploy once, use everywhere. This is the foundation that gives every project professional observability.

Components

Tool	Role	License
Prometheus	Metrics collection & storage (time-series)	Apache 2.0
Grafana	Dashboards, alerting, visualization	AGPL 3.0
Loki	Log aggregation (like CloudWatch Logs but self-hosted)	AGPL 3.0
Promtail	Log shipper (tails files/containers → Loki)	AGPL 3.0
cAdvisor	Container resource metrics (CPU, RAM, network, disk per container)	Apache 2.0
Node Exporter	Host-level metrics (server CPU, RAM, disk, network)	Apache 2.0
PostgreSQL Exporter	Database metrics (connections, queries, cache hit ratio, table sizes)	Apache 2.0
Redis Exporter	Redis metrics (memory, keys, hit/miss ratio, connections)	MIT

Tool	Role	License
Elasticsearch Exporter	ES cluster health, indexing rate, search latency	Apache 2.0
Alertmanager	Alert routing (email via MiaB SMTP, webhook to Listmonk or Slack)	Apache 2.0
Uptime Kuma	Endpoint uptime monitoring with status page	MIT

Architecture



Deployment

- Single `docker-compose.monitoring.yml` at `/opt/monitoring/`
- Prometheus config scrapes all project containers, node exporter, and database exporters
- Promtail tails Docker logs + systemd journal + nginx access/error logs
- Grafana provisioned with dashboards-as-code (JSON files in git)
- Nginx reverse proxy: `monitoring.amiheines.com` (Grafana, auth-protected) and `status.amiheines.com` (Uptime Kuma, public)
- Alertmanager sends alerts via MiaB SMTP to ami@amiheines.com

Grafana Dashboards (provisioned as code)

Dashboard	Content
Server Overview	Host CPU, RAM, disk, network, load average (Node Exporter)
Docker Containers	Per-container CPU, RAM, network I/O, restart count (cAdvisor)
Blockchain Health	Block production rate, head block number, BFT finality, producer status, last irreversible block
ChainVault App	API request rate, error rate, latency p50/p95/p99, active deals, document uploads/day
Verarta App	API request rate, artwork uploads, chunk upload throughput, WebAuthn auth rate
Parking Tickets App	Tickets issued/day, payment conversion rate, expired tickets, revenue
PostgreSQL	Active connections, queries/sec, cache hit ratio, table sizes, slow queries
Redis	Memory usage, key count, hit/miss ratio, connected clients
Elasticsearch	Cluster health, shard count, indexing rate, search latency
Nginx	Request rate, status codes, response times, upstream health
Uptime & SLA	Endpoint availability, response time trends, incident history

Application-Level Metrics (added to each project) Each app exposes a `/metrics` endpoint (Prometheus format) via `prom-client` (Node.js):

Common metrics (all apps): - `http_requests_total{method, route, status}` — request counter - `http_request_duration_seconds{method, route}` — latency histogram - `http_request_size_bytes` / `http_response_size_bytes` — payload sizes - `app_errors_total{type}` — error counter by category - `nodejs_heap_used_bytes` — memory pressure

Blockchain-specific metrics: - `blockchain_head_block_number` — current head block - `blockchain_last_irrever` — finality lag - `blockchain_transactions_total{contract, action}` — on-chain action counter - `blockchain_transaction_duration_seconds` — time to push and confirm a transaction

ChainVault-specific: - `chainvault_deals_total{status}` — deals by status - `chainvault_documents_uploaded_t` — document upload count - `chainvault_wire_verifications_total{result}` — wire fraud check outcomes - `chainvault_stripe_payments_total{status}` — payment outcomes

Verarta-specific: - `verarta_artworks_registered_total` — artwork registration count - `verarta_chunks_uploade` — chunk upload throughput - `verarta_webauthn_auth_total{result}` — biometric auth success/failure - `verarta_quota_usage_ratio{user}` — per-user quota utilization

Parking Tickets-specific: - `parking_tickets_issued_total{city, violation_type}` — tickets by city and type - `parking_tickets_paid_total{city}` — paid tickets - `parking_tickets_expired_total{city}` — expired tickets - `parking_revenue_total_cents{city, currency}` — revenue gauge - `parking_payment_conversi` — paid / (paid + expired) - `parking_image_upload_duration_seconds` — chunked image upload time

Claude Code Development Tasks — Monitoring

#	Task	Est. Time with Claude Code
M1	Create `docker-compose.monitoring.yml` with Prometheus, Grafana, Loki, Promtail, cAdvisor, Node Exporter, Alertmanager	2-3 hours
M2	Write Prometheus scrape configs for all targets	1 hour

#	Task	Est. Time with Claude Code
M3	Write Promtail config (Docker log driver + systemd journal + nginx logs)	1 hour
M4	Create Grafana dashboard JSON files (provisioned as code) — Server, Docker, Blockchain	3-4 hours
M5	Add `prom-client` to ChainVault chain-service + Next.js, expose `/metrics`	2-3 hours
M6	Add `prom-client` to Verarta backend (Astro), expose `/metrics`	2 hours
M7	Create per-app Grafana dashboards (ChainVault, Verarta)	2-3 hours
M8	Add PostgreSQL, Redis, Elasticsearch exporters to compose	1 hour
M9	Create database-layer Grafana dashboards	2 hours
M10	Set up Alertmanager with email alerts (MiaB SMTP)	1 hour
M11	Deploy Uptime Kuma, configure endpoint checks, publish status page	1-2 hours
M12	Nginx configs for monitoring.amiheines.com and status.amiheines.com	30 min

Total monitoring stack: ~20-24 hours with Claude Code (would be 60-80 hours without)

11.2 ChainVault — Expansion

Current state: Production-deployed. Next.js 15, Antelope blockchain (4 BP + SHIP), Hyperion, PostgreSQL, MinIO, Stripe. Wire fraud verification, document vault with client-side AES-256-GCM encryption. Missing: Stripe live keys, observability, some OAuth wiring.

Development Tasks

#	Task	Description	Est. Time
CV1	**Stripe live integration**	Add live keys, test webhook flow end-to-end, verify subscription lifecycle	3-4 hours
CV2	**OAuth SSO completion**	Wire up Google + Microsoft Entra ID login fully, test flow	3-4 hours
CV3	**Prometheus metrics**	Add `prom-client` to chain-service + Next.js API routes, expose `/metrics` with ChainVault-specific counters (see metrics list above)	3-4 hours

#	Task	Description	Est. Time
CV4	**Structured logging**	Replace `console.log` in chain-service and hyperion-sync with `pino` (JSON structured logs). Promtail picks these up automatically via Docker log driver.	2-3 hours
CV5	**Health check dashboard**	Expand `/health` endpoint on chain-service to return detailed status: blockchain connectivity, DB pool, Redis, MinIO, Hyperion. Grafana panel for service health.	2 hours
CV6	**Grafana dashboard**	ChainVault-specific dashboard: API latency, deals created/day, document uploads, wire verifications, Stripe payment outcomes, error rates	3 hours
CV7	**Alert rules**	Alertmanager rules: blockchain stopped producing, API error rate >5%, Stripe webhook failures, disk usage >80%, service restarts	2 hours
CV8	**Smart contract deployment**	Deploy chainvault, auditlog, wirevault contracts to production chain (WASM+ABI). Verify via Hyperion.	2-3 hours
CV9	**Demo data seeder**	Script to populate demo deals, documents, wire verifications for portfolio showcasing	2 hours
CV10	**Public demo mode**	Read-only demo login that lets prospects explore the UI without creating an account. Shows sample deals, documents, audit trail.	3-4 hours

Total ChainVault expansion: ~26-32 hours with Claude Code

11.3 Verarta — Expansion

Current state: Production-deployed. Astro 5 backend, Next.js frontend, Antelope/Spring blockchain (4-node + Savanna consensus), WebAuthn biometric auth, encrypted chunked file uploads, pace controller. Missing: email infrastructure completion, monitoring.

Development Tasks

#	Task	Description	Est. Time
V1	**Email infrastructure**	Complete MiaB integration: verification codes, welcome emails, artwork share notifications. Replace DEV_MODE bypass with actual SMTP sends.	3-4 hours
V2	**Prometheus metrics**	Add `prom-client` to Astro backend, expose `/metrics` with Verarta-specific counters (artworks, chunks, auth, quota)	2-3 hours
V3	**Structured logging**	Replace `console.error()` in all 32 API endpoints with `pino`. Add request ID correlation for tracing.	3-4 hours
V4	**Grafana dashboard**	Verarta-specific dashboard: artwork registrations, upload throughput, WebAuthn success rate, pace controller state, quota utilization	3 hours
V5	**Alert rules**	Blockchain finality lag, pace controller stuck, upload failures, auth failures spike	1-2 hours
V6	**File download/reassembly**	Complete the image retrieval pipeline: query chunks from Hyperion, reassemble, serve with correct MIME type, Redis-cached thumbnails	4-5 hours
V7	**Public gallery**	Public-facing artwork gallery page at verarta.com — shows registered artworks with provenance chain. Great portfolio piece.	4-5 hours
V8	**Multi-device signers**	Allow users to register multiple WebAuthn devices. Device nickname, list/revoke devices. Recovery via trusted contacts (2-of-N threshold).	6-8 hours
V9	**Demo data seeder**	Script to populate demo artworks, users, provenance records for portfolio	2 hours
V10	**Public demo mode**	Read-only exploration of sample artworks, provenance chain, blockchain explorer	3-4 hours

Total Verarta expansion: ~32-40 hours with Claude Code

11.4 Parking Tickets System — New Build

Current state: Detailed plan exists (PARKING_TICKETS_PLAN.md in verarta.com repo). No code written yet. Shares the Verarta blockchain cluster. Full spec includes smart contract, backend API (32 endpoints), frontend (admin + employee + public payment), Stripe payments, on-chain image storage.

This is the strongest portfolio piece because it's a **new vertical** — proves you can apply private blockchain to any domain, not just art or real estate.

Development Tasks

#	Task	Description	Est. Time
P1	**Smart contract**	Write and compile `parking.core` contract (C++, Antelope CDT): cities, employees, price options, tickets, on-chain images. Deploy to `parkingcore` account on Verarta chain.	6-8 hours
P2	**Backend foundation**	Astro 5 SSR project: PostgreSQL schema (8 tables), Redis, auth (JWT), core libs (db, chain, email)	4-5 hours
P3	**Admin auth + city setup**	Registration, email verification, login, city creation (chain + DB), price options CRUD, employee management	5-6 hours
P4	**Ticket issuance**	Employee endpoint: create ticket on chain + DB, vehicle registry lookup (mock), GPS/location handling	4-5 hours
P5	**On-chain image upload**	Chunked image upload (mirrors Verarta pattern): init → chunk upload → complete. Image retrieval and thumbnail generation.	5-6 hours
P6	**Stripe payment**	PaymentIntent creation, webhook handler, ticket status update on chain, confirmation emails	4-5 hours
P7	**Deadline expiration**	node-cron job: expire open tickets past deadline, chain action, email notification	2 hours
P8	**Frontend — Admin**	Next.js 16: admin dashboard (stats, ticket list, filters, employee mgmt, price options), city setup wizard	8-10 hours

#	Task	Description	Est. Time
P9	**Frontend — Employee**	Mobile-first: new ticket form (camera, GPS, plate input, price selector), ticket list, image upload progress	6-8 hours
P10	**Frontend — Public payment**	Public ticket view page, Stripe Elements payment form, success/failure	3-4 hours
P11	**Email templates**	Ticket issued, payment confirmation, deadline reminder, expired notice	2-3 hours
P12	**Prometheus metrics**	`prom-client` in Astro backend: tickets issued, paid, expired, revenue, image upload duration	2-3 hours
P13	**Grafana dashboard**	Parking-specific: tickets/day by city, payment conversion rate, revenue, violation type breakdown, employee productivity	3 hours
P14	**Alert rules**	Payment failures, expiration job missed, high error rate, blockchain lag	1-2 hours
P15	**Demo data seeder**	Populate demo city, employees, tickets (open/paid/expired), images for portfolio	2-3 hours
P16	**Deployment**	Nginx config (parking.amiheines.com), PM2 ecosystem, production env, deploy script	2-3 hours

Total Parking Tickets system: ~55-72 hours with Claude Code (would be 150-200 hours without)

11.5 Development Schedule

Assuming 20 hours/week, with Claude Code acceleration. Outreach and content continue in parallel (the other 30% of time from section 4).

Week	Focus	Tasks	Deliverable
1-2	Monitoring stack	M1-M6	Prometheus + Grafana + Loki running, ChainVault + Verarta metrics exposed
3	Monitoring polish + ChainVault	M7-M12, CV1-CV2	All dashboards live, Uptime Kuma public, Stripe + OAuth live on ChainVault
4	ChainVault expansion	CV3-CV7	ChainVault fully monitored with alerts

Week	Focus	Tasks	Deliverable
5	ChainVault finish + Verarta start	CV8-CV10, V1-V2	ChainVault demo mode live, Verarta email + metrics
6-7	Verarta expansion	V3-V8	Verarta fully monitored, file download working, public gallery, multi-device
8	Verarta finish + Parking start	V9-V10, P1-P2	Verarta demo mode, Parking contract + backend foundation
9-10	Parking core features	P3-P7	Admin flow, ticketing, images, payments, expiration all working
11-12	Parking frontend	P8-P11	Full UI: admin dashboard, employee mobile, public payment
13	Parking monitoring + deploy	P12-P16	Parking fully monitored, deployed, demo data populated
14	Integration + portfolio	Cross-project	All three apps live with monitoring, demo modes, seeded data. Portfolio page on amiheines.com updated with live links + screenshots.

Total: ~14 weeks (3.5 months) for all three projects fully built, monitored, and demo-ready.

11.6 Portfolio Value — What Prospects See

When a prospect visits amiheines.com/portfolio or joins a discovery call, they see:

What they see	What it proves
3 live blockchain apps across different verticals (real estate, art, municipal)	"He can apply this to MY industry"
Grafana dashboards with real metrics on every app	"He ships production-grade, not prototypes"
Public status page (status.amiheines.com) with uptime history	"This runs reliably — I can trust my data to this"
Demo mode — click and explore without signing up	"I can see exactly what I'd get"
One server running everything: blockchain, 3 apps, monitoring, email, analytics	"He can do this on our infrastructure too"
Open-source stack with zero licensing fees	"No vendor lock-in, no surprise costs"
Blog/video content showing how each feature was built	"He's transparent about the process"
Built with Claude Code — AI-accelerated delivery	"He ships fast — my $5K PoC won't take 6 months"

11.7 Content Tie-In — Videos From Development

Every development task above is a potential "Build With Me" video or blog post. High-value examples:

Dev Task	Video/Blog Title
M1-M4	"Enterprise Monitoring for $0: Grafana + Prometheus on a Single Server"

Dev Task	Video/Blog Title
M5-M7	“Adding Observability to a Blockchain App in 3 Hours with Claude Code”
CV3-CV6	“Monitoring a Private Blockchain: What Metrics Actually Matter”
CV10	“Building a Demo Mode So Prospects Can Try Before They Buy”
V6-V7	“Reassembling Files Stored on a Blockchain — The Art Provenance Pipeline”
V8	“Multi-Device WebAuthn for Blockchain Accounts: A Recovery Strategy”
P1	“Writing an Antelope Smart Contract for Parking Tickets — From Scratch”
P5	“Storing Evidence Photos on a Blockchain: Chunked Upload Deep Dive”
P6	“Stripe + Blockchain: Marking a Parking Ticket Paid On-Chain”
P8-P9	“Building a Mobile-First Ticket Issuing App with Next.js and Camera API”
P12-P13	“Grafana Dashboard for a Municipal Parking System — Real-Time Revenue Tracking”
All	“How I Built 3 Blockchain Apps in 14 Weeks Using Claude Code”

11.8 Open-Source Tools Reference Card

This is the pitch to prospects: “Everything here runs on free, battle-tested open-source software.”

Category	Tool	Replaces (Paid)	Annual Cost Saved
Blockchain	Antelope/Spring	Hyperledger managed service, Azure Blockchain	$5-50K
Monitoring	Prometheus + Grafana	Datadog, New Relic	$15-60K
Log aggregation	Loki + Promtail	Splunk, Datadog Logs	$10-40K
Uptime monitoring	Uptime Kuma	PagerDuty, Pingdom	$1-5K
Email marketing	Listmonk	Mailchimp, ConvertKit	$1-5K
Analytics	Matomo	Google Analytics 360	$2-10K
Email server	Mail-in-a-Box	Google Workspace, Exchange	$1-3K
Object storage	MinIO	AWS S3	$1-10K
Search/indexing	Elasticsearch	Algolia, managed ES	$3-15K
Container orchestration	Docker Compose	ECS, EKS, managed K8s	$2-10K
SSL	Let’s Encrypt + Certbot	Commercial SSL certs	$0.5-2K
Reverse proxy	Nginx	AWS ALB, Cloudflare	$1-5K
Process manager	PM2	Managed app hosting	$1-3K
Comment system	ChainTalk (custom)	Disqus, Commento	$1-5K
		Total annual savings:	**$40-200K+**

11.9 POFOV — Long-Tail SEO Lead Generation Engine

Current state: pofov.com is a live Astro 5 site with 1,000+ cryptocurrency project pages, powered by a Python FastAPI scraper backend (scrapecmc.pofov.com) that collects listings, deep FAQ data, and AI-generated Q&A from multiple LLM providers (Gemini, Claude Haiku, GPT-4o-mini). It has Matomo tracking, a contact form

with email notifications, and categorization across 17 verticals. What it **doesn't** have: email capture, content gating, a funnel to amiheines.com, or blockchain-specific educational content targeted at decision-makers.

The opportunity: POFOV already has massive long-tail SEO surface area — 1,000+ pages targeting searches like "what is [coin] good for", "does [coin] support smart contracts", "[coin] blockchain type". The problem is that this traffic currently has no conversion path. Visitors read, leave, and never come back.

The strategy: Turn POFOV from a passive research tool into an **active lead generation engine** that: 1. Captures emails from blockchain-curious visitors 2. Delivers immediate value (tools, comparisons, guides) in exchange 3. Nurtures toward amiheines.com consulting via content and Listmonk sequences 4. Cross-promotes the blockchain comment system (section 11.10) as a live demo

11.9.1 Email Capture — Value-First Gating **Principle:** The first visit is always free. Gate the **second level of value** behind email.

Free (no email)	Gated (email required)	How it drives leads
Basic coin page: about, whitepaper summary, AI Q&A	**Comparison tool:** side-by-side comparison of 2-3 coins (tech specs, use cases, consensus, storage)	"Compare Hyperledger vs Corda vs Antelope" — exactly the search a CTO does before choosing
Category browsing (AI, DeFi, Enterprise, etc.)	**Enterprise Blockchain Buyer's Guide** (PDF): private vs public, consensus comparison, storage capabilities, cost analysis	Direct lead magnet for consulting prospects
Single coin AI answers	**Full AI Research Report** (all 5 questions, all 3 providers, with analysis) as downloadable PDF per coin	Converts researchers into email subscribers
Basic stats (rank, price, market cap)	**Blockchain Capability Matrix:** filterable table of which chains support smart contracts, on-chain storage, privacy, enterprise features	High-value tool that CTOs bookmark and share
17-category grid view	**Weekly Blockchain Intelligence Digest** (Listmonk newsletter): new coins, category trends, enterprise adoption signals	Recurring touchpoint for nurturing

11.9.2 Funnel Architecture

```
Long-tail Google search
  ("what blockchain supports on-chain storage")
        │
        ▼
   pofov.com/project/{slug}
   (free: AI Q&A, about, stats)
        │
        ├──▶ "Compare with other chains" → email gate → comparison tool
        ├──▶ "Download full report" → email gate → PDF report
        ├──▶ "Enterprise Buyer's Guide" → email gate → PDF guide
        └──▶ Banner: "Need a private blockchain for YOUR business?"
                → link to amiheines.com/services
        │
        ▼ (after email capture)
   Listmonk welcome sequence (POFOV-specific, 5 emails):
     1. Deliver the gated content + "here's what else POFOV offers"
     2. "3 things most teams get wrong choosing a blockchain platform"
     3. "Private vs public blockchain: when each one wins" (links to blog)
     4. Case study: "How ChainVault uses Antelope for wire fraud prevention"
     5. "Want help evaluating blockchain for your project?" → amiheines.com/contact
```

```
Ongoing: Weekly Blockchain Intelligence Digest
  → every 4th issue includes consulting CTA
```

11.9.3 Cross-Promotion to amiheines.com Every POFOV page gets:

- **Top banner (subtle):** "Built by Ami Heines — Blockchain Infrastructure Consultant" → links to amiheines.com
- **Enterprise section highlight:** The 7 enterprise blockchain pages (Hyperledger, Antelope, Corda, Quorum, etc.) get a prominent CTA: "Need help deploying [platform] in your organization? → Book a free 30-min discovery call"
- **Footer:** "POFOV is a project by Ami Heines. I help companies build tamper-proof infrastructure with private blockchains. → Learn more"
- **Comparison tool results page:** After comparing chains, show: "Not sure which is right for your use case? I can help. → $500 architecture review"
- **Blog post cross-links:** POFOV blog posts on blockchain comparisons link to amiheines.com blog posts on implementation

11.9.4 New Content for POFOV (SEO + Lead Gen) **New page types to add:**

Page	URL	SEO Target	Gated?
Comparison pages	`/compare/{slug1}-vs-{slug2}`	"hyperledger vs corda", "antelope vs quorum"	Partial — summary free, full comparison gated
Category deep dives	`/category/{name}`	"best enterprise blockchain 2026", "blockchain for healthcare"	Free, with gated PDF download
Capability matrix	`/matrix`	"blockchain comparison chart", "which blockchain supports smart contracts"	Free to view, email to export/filter
Blog/guides	`/guides/{slug}`	"how to choose a private blockchain", "blockchain storage comparison"	Free, with gated extended version
Tools	`/tools/storage-calculator`	"blockchain storage cost calculator"	Free to use, email to save results
Enterprise landing pages	`/enterprise/{slug}`	"hyperledger fabric consulting", "antelope blockchain development"	Free, strong CTA to amiheines.com

High-value comparison pages to create first (highest search intent): 1. Hyperledger Fabric vs Antelope 2. Hyperledger Fabric vs R3 Corda 3. Antelope vs ConsenSys Quorum 4. Private blockchain vs permissioned database 5. Hyperledger Fabric vs ConsenSys Quorum 6. On-chain storage: which blockchains support it? 7. Blockchain consensus mechanisms compared (BFT, PBFT, Raft, Savanna)

11.9.5 POFOV Development Tasks

#	Task	Description	Est. Time
PF1	**Email capture infrastructure**	Listmonk API integration on pofov.com: signup modals, inline forms, double opt-in. Create POFOV-specific list in Listmonk.	3-4 hours

#	Task	Description	Est. Time
PF2	**Content gating system**	Middleware that checks email cookie. Ungated = free preview + email wall. After signup, full content unlocked. Store state in cookie + Listmonk subscriber status.	4-5 hours
PF3	**Comparison tool**	New `/compare/{slug1}-vs-{slug2}` pages. Pull data from both coins, render side-by-side table (consensus, storage, smart contracts, speed, use cases, AI analysis). Auto-generate for top 20 enterprise pair combinations.	6-8 hours
PF4	**PDF report generator**	Server-side PDF generation (puppeteer or @react-pdf/renderer): per-coin full AI report with all provider answers, formatted with POFOV branding. Served after email capture.	4-5 hours
PF5	**Capability matrix page**	`/matrix` — filterable, sortable table of blockchain capabilities. Data derived from AI scrape answers + manual curation for enterprise chains. Export to CSV (gated).	4-5 hours
PF6	**Enterprise landing pages**	Enhanced pages for the 7 enterprise blockchains with consulting CTAs, implementation guides, and links to amiheines.com/services.	3-4 hours
PF7	**Cross-promotion banners**	Top banner, footer, and contextual CTAs linking to amiheines.com across all POFOV pages. A/B testable via Matomo.	2-3 hours
PF8	**Listmonk sequences**	POFOV-specific welcome sequence (5 emails) + weekly digest template. Configure in Listmonk.	3-4 hours
PF9	**Category deep dive pages**	`/category/{name}` — editorial-quality pages for each of the 17 categories with curated picks, trends, and downloadable PDF summaries.	5-6 hours

#	Task	Description	Est. Time
PF10	**Monitoring**	Add `prom-client` to pofov.com Astro backend + scraper FastAPI. Grafana dashboard: page views by coin, email signups/day, comparison tool usage, API scrape job health.	3-4 hours
PF11	**Scraper improvements**	Add scheduling (cron) for weekly auto-rescrape. Add new AI questions targeting enterprise use cases. Add chain capability classification (storage, smart contracts, privacy, speed).	4-5 hours
PF12	**ChainTalk integration**	Add the blockchain comment system (section 11.10) to POFOV coin pages — visitors discuss coins with on-chain comments. First live demo of ChainTalk.	3-4 hours

Total POFOV expansion: ~45-57 hours with Claude Code

11.9.6 Why This Works for Revenue POFOV doesn't generate revenue directly. It generates **leads at scale** with zero marginal cost:

Metric	Conservative (month 3)	Optimistic (month 6)
Organic search visitors/month	2,000	10,000
Email capture rate	3%	5%
New emails/month	60	500
Consulting-qualified leads (enterprise interest)	3-5	15-25
Discovery calls booked	1-2	3-5
Revenue from POFOV-sourced leads	$500-$1,500/month	$2,500-$7,500/month

The 1,000+ pages are a **moat** — no competitor will hand-build this. The AI-generated content + scraper pipeline makes it maintainable with weekly auto-rescrapes.

11.10 ChainTalk — Blockchain-Based Embeddable Comment System

Concept: A drop-in comment/chat widget that any website can add with a single `<script>` tag. Comments are stored on the Antelope/Spring blockchain — immutable, tamper-proof, censorship-resistant, and verifiable. This is the **simplest possible way** for someone to experience blockchain value on their own website.

Why this is a killer portfolio piece: - It's the most **relatable** demo — everyone understands comments - It's **embeddable** — prospects can try it on their own site in 5 minutes - It **sells itself** — "Your comments can't be silently deleted, edited, or backdated. Every message has a blockchain receipt." - It creates **network effects** — every site that embeds it sends traffic back to you - It's a **Trojan horse** for consulting — site owners who embed ChainTalk start asking "what else can blockchain do for my business?"

11.10.1 Architecture

```
┌──────────────────────────────────────────────────────────┐
│        Any Website (embed)                               │
│                                                          │
│  <script src="chaintalk.js"></script>                    │
│  <div id="chaintalk"                                     │
│       data-site="example.com"                            │
│       data-page="/blog/post-1">                          │
│  </div>                                                  │
│                                                          │
│   ┌──────────────────────────────────────────────┐       │
│   │  ChainTalk Widget (iframe)                   │       │
│   │                                              │       │
│   │   ┌──────────────────────────────┐           │       │
│   │   │ Comment 1 -- @alice          │           │       │
│   │   │ "Great post!" [chain]        │           │       │
│   │   │ Comment 2 -- @bob            │           │       │
│   │   │ "Disagree..."[chain]         │           │       │
│   │   │                              │           │       │
│   │   │ [Write a comment...]         │           │       │
│   │   │ [Post to blockchain]         │           │       │
│   │   └──────────────────────────────┘           │       │
│   └──────────────────────────────────────────────┘       │
└──────────────────────────────────────────────────────────┘
            │                  ▲
            ▼                  │
┌──────────────────────────────────────────────────────────┐
│  ChainTalk Backend                                       │
│  chaintalk.amiheines.com                                 │
│                                                          │
│  Astro 5 SSR + Node.js                                   │
│   ┌──────────────────┐   ┌──────────────────┐           │
│   │ REST API         │   │ PostgreSQL       │           │
│   │ /api/comments    │   │ (read cache)     │           │
│   │ /api/auth        │   │                  │           │
│   │ /api/sites       │   │                  │           │
│   └──────────────────┘   └──────────────────┘           │
│          │                                               │
│          ▼                                               │
│   ┌──────────────────────────────────────────────┐       │
│   │ Antelope/Spring Blockchain                   │       │
│   │ (shared Verarta cluster)                     │       │
│   │                                              │       │
│   │ Contract: chaintalk.core                     │       │
│   │ Account:  chaintalkco                        │       │
│   │                                              │       │
│   │ Tables:                                      │       │
│   │ - sites (registered domains)                 │       │
│   │ - comments (immutable)                       │       │
│   │ - users (accounts)                           │       │
│   │ - votes (upvote/downvote)                    │       │
│   │ - moderations (flag, hide)                   │       │
│   └──────────────────────────────────────────────┘       │
└──────────────────────────────────────────────────────────┘
```

11.10.2 Smart Contract — `chaintalk.core` Deployed to `chaintalkco` account on the shared Verarta blockchain.

Tables:

Table	Fields	Purpose
`sites`	site_id, owner (name), domain, display_name, created_at, is_active, settings (moderation mode, theme)	Registered website domains
`comments`	comment_id, site_id, page_path, author (name), parent_id (0 for root, >0 for reply), body (text, max 2000 chars), body_hash (sha256), created_at, is_hidden	Immutable comment storage
`users`	user_id, account (name), display_name, email_hash, avatar_url, created_at	User profiles
`votes`	vote_id, comment_id, voter (name), value (+1/-1), created_at	Upvotes/downvotes
`moderations`	mod_id, comment_id, moderator (name), action (hide/unhide/flag), reason, created_at	Moderation audit trail — comments are never deleted, only hidden

Key design decisions: - **Comments are never deleted from chain** — only hidden via moderation action. The original text remains on-chain forever. This is the core value proposition: "Your comment history is permanent and auditable." - **Moderation is transparent** — every hide/unhide is a blockchain transaction with a reason. Site owners can moderate, but the act of moderation is itself recorded. - **Replies are threaded** — `parent_id` creates a tree structure. Frontend renders nested threads. - **Votes are on-chain** — no vote manipulation possible.

Actions:

Action	Auth	Description
`regsite`	Self	Register a new site (domain verification via DNS TXT record or meta tag)
`updatesite`	Site owner	Update site settings (moderation mode, theme)
`reguser`	Self	Register a user account
`postcomment`	User	Post a comment or reply to a page on a registered site
`editcomment`	Author	Edit comment body (creates new version — old version preserved in blockchain history via Hyperion)
`vote`	User	Upvote or downvote a comment (one vote per user per comment)
`moderate`	Site owner	Hide/unhide/flag a comment (with reason)

11.10.3 Embed Widget

Integration for site owners (as simple as Disqus):

```
<!-- Add to any page -->
<script src="https://chaintalk.amiheines.com/embed.js" async></script>
<div id="chaintalk"
     data-site-id="your-site-id"
     data-page-id="/blog/my-post"
     data-theme="light">
</div>
```

Widget features: - **Iframe-based** — isolated CSS, no style conflicts with host site - **Lightweight** — <30KB JS bundle (vanilla JS, no framework in embed loader) - **Responsive** — works on mobile and desktop - **Theming** — light/dark mode, customizable accent color via `data-theme` and CSS variables - **Auth options:** Email/password,

or anonymous with display name (site owner configures) - **Blockchain receipt** — every comment shows a small chain icon linking to the blockchain transaction on the Verarta explorer - **Threaded replies** — nested conversation threads - **Voting** — upvote/downvote with live counts - **Moderation badge** — if a comment was hidden by moderator, shows "[hidden by site owner — reason: …]" with a link to the moderation transaction - **"Powered by ChainTalk"** — footer link back to chaintalk.amiheines.com (free tier)

11.10.4 Backend API

Method	Route	Auth	Description
POST	`/api/sites/register`	API key	Register a new site domain
GET	`/api/sites/:id/verify`	API key	Check DNS/meta-tag domain verification
GET	`/api/comments/:siteId/:pageId`	Public	Get comments for a page (paginated, threaded)
POST	`/api/comments`	User	Post a comment (pushes to blockchain)
PUT	`/api/comments/:id`	Author	Edit a comment (new chain transaction)
POST	`/api/comments/:id/vote`	User	Vote on a comment
POST	`/api/comments/:id/moderate`	Site owner	Hide/unhide with reason
POST	`/api/auth/register`	Public	Create user account (email + password)
POST	`/api/auth/login`	Public	Login, get JWT
GET	`/api/auth/session`	User	Get current session
GET	`/api/embed.js`	Public	Serve the embed script

11.10.5 Why Blockchain Matters Here (The Pitch)

This is the slide you show in every discovery call:

Traditional Comments (Disqus, etc.)	ChainTalk (Blockchain)
Platform can delete any comment silently	Comments are **permanent** — hidden, never deleted
Edits overwrite history — no trace	Edit history preserved in blockchain — every version auditable
Vote counts can be manipulated server-side	Votes are on-chain — **mathematically verifiable**
Moderation is opaque — users don't know why	Moderation actions are **transparent** — recorded on-chain with reason
Platform goes down = comments lost forever	Blockchain cluster keeps running — **data survives the platform**
No proof when a comment was posted	**Cryptographic timestamp** — provable posting time
Company can retroactively edit user content	**Immutable** — not even the site owner can alter past comments

One-liner for prospects: "If you wouldn't trust your sysadmin to never edit a database row, you need blockchain. ChainTalk is the simplest proof of that principle."

11.10.6 Free Tier & Upsell

Tier	Price	Limits	Features
Free	$0	1 site, 500 comments/month, "Powered by ChainTalk" badge	Full blockchain storage, basic moderation, light/dark theme
Pro	$29/month	5 sites, 5,000 comments/month, no badge	Custom CSS, priority API, email notifications on new comments

Tier	Price	Limits	Features
Enterprise	$99/month or custom	Unlimited sites, unlimited comments	Custom branding, webhook integrations, dedicated support, SLA
Self-hosted	$2,500 one-time	Deploy on your own blockchain	Full source code, your infrastructure, Ami sets it up for you

The self-hosted tier is the **consulting Trojan horse**: "Want ChainTalk on your own private blockchain? That's a $2,500 setup + $500/month managed infra. Oh, and while we're setting up your chain, let's talk about what else you could put on it…"

11.10.7 Development Tasks

#	Task	Description	Est. Time
CT1	**Smart contract**	Write and compile `chaintalk.core` (C++, Antelope CDT): sites, comments, users, votes, moderations tables and actions. Deploy to `chaintalkco` on Verarta chain.	5-6 hours
CT2	**Backend foundation**	Astro 5 SSR: PostgreSQL schema (comment read cache, sites, users), Redis, JWT auth, chain integration lib	4-5 hours
CT3	**Comment CRUD API**	GET comments (paginated, threaded tree), POST comment (chain + DB), PUT edit (chain + DB), vote, moderate	5-6 hours
CT4	**Site registration + verification**	Register site, DNS TXT or meta-tag domain verification, API key generation	3-4 hours
CT5	**User auth**	Email/password registration, login, JWT sessions. Anonymous posting option (configurable per site).	3-4 hours
CT6	**Embed widget — JS loader**	Vanilla JS embed script (<30KB): creates iframe, handles resize, passes config, postMessage communication between iframe and host page	4-5 hours
CT7	**Embed widget — iframe UI**	Comment list (threaded), post form, login/register, vote buttons, blockchain receipt links, moderation badges. Responsive. Light/dark theme.	6-8 hours

#	Task	Description	Est. Time
CT8	**ChainTalk landing page**	Marketing page at chaintalk.amiheines.com: what it is, how it works, live demo, pricing, “Add to your site in 2 minutes” setup wizard	4-5 hours
CT9	**Admin dashboard**	Site owner dashboard: comment stats, moderation queue, API key management, embed code generator, theme customizer	5-6 hours
CT10	**Prometheus metrics**	`prom-client`: comments posted/day, active sites, auth events, blockchain transaction latency, cache hit rate	2-3 hours
CT11	**Grafana dashboard**	ChainTalk-specific: comments/day, sites registered, votes, moderation actions, embed loads, API latency	2-3 hours
CT12	**Deployment**	Nginx config (chaintalk.ami-heines.com), PM2 ecosystem, production env, deploy script	2-3 hours
CT13	**Integrate into POFOV**	Add ChainTalk embed to all pofov.com coin pages — first live deployment, creates discussion under every coin	3-4 hours
CT14	**Integrate into amiheines.com**	Add ChainTalk to blog posts on amiheines.com — eat your own dogfood	1-2 hours
CT15	**Demo site**	Create a sample blog at demo.chaintalk.ami-heines.com with 5 posts and seeded comments showing threading, voting, moderation — the “try it live” experience	3-4 hours

Total ChainTalk system: ~53-68 hours with Claude Code (would be 150-200 hours without)

11.10.8 Content Tie-In

Dev Task	Video/Blog Title
CT1	“Writing a Smart Contract for a Comment System — Why Blockchain Comments Can’t Be Silently Deleted”
CT6-CT7	“Building an Embeddable Widget with Vanilla JS and iframes — No React Required”
CT5	“Blockchain-Backed Authentication: When Signing a Comment Means Signing a Transaction”
CT13	“Adding Blockchain Comments to 1,000 Pages in One Deploy”

Dev Task	Video/Blog Title
All	"I Built a Blockchain Alternative to Disqus — Here's Why and How"
Pitch	"Why Your Comment Section Should Be on a Blockchain (5 Minutes for CTOs)"

11.11 Updated Development Schedule

Revised to include POFOV and ChainTalk alongside the original projects.

Week	Focus	Tasks	Deliverable
1-2	Monitoring stack	M1-M6	Prometheus + Grafana + Loki running, metrics exposed
3	Monitoring polish + ChainVault	M7-M12, CV1-CV2	Dashboards live, Uptime Kuma, Stripe + OAuth on ChainVault
4	ChainVault expansion	CV3-CV7	ChainVault fully monitored with alerts
5	ChainVault finish + Verarta start	CV8-CV10, V1-V2	ChainVault demo mode, Verarta email + metrics
6-7	Verarta expansion	V3-V8	Verarta monitored, gallery, multi-device
8	Verarta finish + Parking start	V9-V10, P1-P2	Verarta demo mode, Parking contract + backend
9-10	Parking core features	P3-P7	Ticketing, images, payments, expiration
11-12	Parking frontend + deploy	P8-P11, P12-P16	Full UI, monitoring, deployed
13-14	POFOV lead gen	PF1-PF6	Email capture, gating, comparison tool, PDF reports, matrix
15	POFOV finish	PF7-PF11	CTAs, Listmonk sequences, category pages, monitoring, scraper upgrades
16-17	ChainTalk core	CT1-CT5	Smart contract, backend, comments API, auth, site registration
18-19	ChainTalk widget + landing	CT6-CT9	Embed widget, landing page, admin dashboard
20	ChainTalk deploy + integrate	CT10-CT15	Monitoring, deploy, embed on POFOV + amiheines.com + demo site
21	Integration + portfolio	Cross-project	All 5 projects live. Portfolio page updated. Public demos. Status page.

Total: ~21 weeks (5 months) for all five projects fully built, monitored, and demo-ready. Total estimated dev hours: ~230-290 hours with Claude Code (would be 600-800 hours without)

11.12 Updated Portfolio Value — What Prospects See

What they see	What it proves
5 live blockchain apps across different verticals	“He can apply this to literally anything”
1,000+ page SEO site (POFOV) generating organic traffic	“He understands marketing AND engineering”
Embeddable widget they can try on their own site in 2 minutes	“I can experience the value before paying a cent”
Grafana dashboards with real metrics on every app	“Production-grade, not prototypes”
Public status page with uptime history	“Reliable infrastructure”
Blockchain comment discussions under every POFOV coin page	“Blockchain can be invisible and useful, not just hype”
One server running 5 apps + monitoring + email + analytics	“He can do this on our infrastructure”
Open-source everything with zero licensing fees	“No vendor lock-in”
Built with Claude Code — AI-accelerated delivery	“He ships fast”

Week 1 — Foundation: Website + Outreach Setup + Monitoring Start

Dates: _____ to _____ **Focus:** 70% business foundation / 30% monitoring infra start **Target hours:** 20

Business Foundation (14 hours)

amiheines.com Website Launch (8 hours)

- ☐ Fork/adapt the Dante Astro theme from privateblockchaindatabase.com
- ☐ Create landing page: hero section, value prop, social proof placeholder
- ☐ Create /about page: bio, background, philosophy
- ☐ Create /services page with all 5 tiers ($500 → $5K → managed infra)
- ☐ Create /portfolio page: ChainVault and Verarta case studies with screenshots
- ☐ Create /contact page: contact form + Cal.com booking embed
- ☐ Create /newsletter page: Listmonk signup form
- ☐ Create /resources page: gated lead magnets (placeholder for now)
- ☐ Create /blog page: migrate 16 blog posts from privateblockchaindatabase.com
- ☐ Integrate Listmonk API for newsletter signup (test with real subscription)
- ☐ Add Matomo tracking to all pages
- ☐ Deploy to server (158.69.27.200): nginx config, certbot SSL
- ☐ Set up 301 redirects from privateblockchaindatabase.com → amiheines.com/blog
- ☐ **Verify:** visit every page, test signup flow, confirm Matomo fires, check redirects

LinkedIn Profile Optimization (2 hours)

- ☐ Rewrite headline: problem you solve, not job title (e.g., "I build tamper-proof infrastructure for companies that can't afford data tampering — Private Blockchain | Wire Fraud Prevention | Audit Trails")
- ☐ Update About section: who you help, what you've built, CTA to book a call
- ☐ Add amiheines.com as website
- ☐ Add ChainVault and Verarta as projects/experience
- ☐ Update banner image (professional, relates to blockchain infra)
- ☐ Connect with 20-30 people in target verticals (don't pitch yet — just connect)

Lead Magnet — First Draft (2 hours)

- ☐ Write "Private Blockchain Infrastructure Checklist" (PDF, 2-3 pages)
  - What to evaluate before choosing a platform
  - Key questions for your CTO/VP Engineering
  - Red flags in vendor proposals
- ☐ Design in simple template (Canva or Markdown → PDF)
- ☐ Upload to /resources page, gate behind Listmonk email capture

Freelance Platform Profiles (2 hours)

- ☐ Create/update Toptal profile (apply if not yet accepted)
- ☐ Create/update Upwork profile (expert-vetted tier)
- ☐ Create/update Catalant profile
- ☐ Position as: blockchain infrastructure / tamper-proof records specialist
- ☐ Include links to amiheines.com, ChainVault, Verarta as portfolio

---

Monitoring Stack Start (6 hours)

M1 — docker-compose.monitoring.yml (3-4 hours)

- ☐ Create `/opt/monitoring/` directory structure on server
- ☐ Write `docker-compose.monitoring.yml` with:

    - Prometheus (latest)
    - Grafana (latest, provisioned with admin password)
    - Loki (latest)
    - Promtail (latest)
    - cAdvisor (latest)
    - Node Exporter (latest)
    - Alertmanager (latest)
- ☐ Configure volume mounts for persistent data
- ☐ Configure networking (all on internal Docker network)
- ☐ Test: `docker compose up -d`, verify all containers healthy

M2 — Prometheus scrape configs (1 hour)

- ☐ Write `prometheus.yml` with scrape targets:
    - Node Exporter (host metrics)
    - cAdvisor (container metrics)
    - Prometheus self-monitoring
    - Grafana health
- ☐ Set scrape intervals (15s for critical, 60s for others)
- ☐ Test: Prometheus UI shows all targets as UP

M3 — Promtail config (1 hour)

- ☐ Write `promtail.yml`:
    - Docker container log discovery (auto-discover all containers)
    - systemd journal (chainvault-web, chainvault-chain-service, etc.)
    - nginx access + error logs (/var/log/nginx/)
- ☐ Label enrichment: container_name, service, log_level
- ☐ Test: Grafana Explore → Loki → can see container logs

End-of-Week Checkpoints

- ☐ amiheines.com is live with all pages, working forms, Matomo tracking
- ☐ 301 redirects from privateblockchaindatabase.com working
- ☐ LinkedIn profile optimized and 20+ new connections
- ☐ Lead magnet PDF created and gated on /resources
- ☐ Freelance platform profiles submitted
- ☐ Prometheus + Grafana + Loki running on server
- ☐ Can view host metrics and container logs in Grafana

Week 2 — Outreach Launch + Listmonk Setup + Monitoring Dashboards

Dates: _____ to _____ **Focus:** 50% outreach & email infrastructure / 50% monitoring completion **Target hours:** 20

Outreach & Email (10 hours)

Prospect List Building (4 hours)

- ☐ Build spreadsheet/CRM with 120+ contacts across 4 verticals:
  - 50 real estate title company CTOs / IT directors
  - 30 legal firm technology partners / CIOs
  - 20 healthcare compliance officers
  - 20 government IT directors
- ☐ For each: name, company, title, LinkedIn URL, specific pain point note
- ☐ Research 5-10 companies deeply (recent news, transactions, compliance issues) for personalized outreach

Outreach Templates (2 hours)

- ☐ Write 3-4 LinkedIn message templates:
  - Template A: Real estate / wire fraud angle
  - Template B: Legal / document integrity angle
  - Template C: Healthcare / audit trail angle
  - Template D: General tamper-proof records angle
- ☐ Each template: 1 specific reference to their company/industry, 1 pain point sentence, offer free 30-min audit, link to amiheines.com
- ☐ Create discovery call script (30 min format: 10 min listen, 10 min diagnose, 10 min propose)
- ☐ Create proposal template (1-page, covers all tiers)

First Outreach Batch (2 hours)

- ☐ Send 15-20 personalized LinkedIn messages (use templates, customize per prospect)
- ☐ Comment on 10-15 target prospects' LinkedIn posts (add value, don't pitch)
- ☐ Post first LinkedIn insight (1-2 paragraphs on wire fraud, audit trails, or tamper-proof records)
- ☐ Track: messages sent, connection requests accepted, responses received

Listmonk Setup — Phase 3 (2 hours)

- ☐ Create lists in Listmonk:
  - amiheines.com subscribers
  - POFOV subscribers (for later)
  - Lead magnet downloaders
- ☐ Configure double opt-in
- ☐ Create email templates (branded, responsive HTML)
- ☐ Build welcome sequence (5 emails over 14 days):
  1. Welcome + link to best resource (checklist PDF)
  2. Your story + why private blockchain
  3. Case study: ChainVault
  4. Common misconceptions about blockchain in enterprise
  5. "How I can help" + book a call CTA
- ☐ Test: sign up → receive welcome email → sequence triggers correctly

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Monitoring Stack Completion (10 hours)

M4 — Grafana Dashboard JSON Files (3-4 hours)

- ☐ Create Server Overview dashboard: Host CPU, RAM, disk, network, load average
- ☐ Create Docker Containers dashboard: per-container CPU, RAM, network I/O, restart count
- ☐ Create Blockchain Health dashboard: block production rate, head block, finality
- ☐ Provision all dashboards as code (JSON files in `/opt/monitoring/grafana/dashboards/`)
- ☐ Configure Grafana datasources provisioning (Prometheus + Loki)
- ☐ Test: dashboards auto-load on fresh Grafana start

M5 — ChainVault Prometheus Metrics (2-3 hours)

- ☐ Install `prom-client` in ChainVault chain-service
- ☐ Add metrics middleware to Express.js:
  - `http_requests_total{method, route, status}`
  - `http_request_duration_seconds{method, route}`
  - `blockchain_head_block_number`
  - `blockchain_transactions_total{contract, action}`
  - `chainvault_deals_total{status}`
  - `chainvault_documents_uploaded_total`
- ☐ Expose `GET /metrics` endpoint on chain-service
- ☐ Add `prom-client` to Next.js API routes (via middleware)
- ☐ Add ChainVault scrape targets to Prometheus config
- ☐ Test: Prometheus can scrape ChainVault metrics

M6 — Verarta Prometheus Metrics (2 hours)

- ☐ Install `prom-client` in Verarta Astro backend
- ☐ Add metrics middleware:
  - `http_requests_total{method, route, status}`
  - `http_request_duration_seconds{method, route}`
  - `verarta_artworks_registered_total`
  - `verarta_chunks_uploaded_total`
  - `verarta_webauthn_auth_total{result}`
- ☐ Expose `GET /metrics` endpoint
- ☐ Add Verarta scrape targets to Prometheus config
- ☐ Test: Prometheus can scrape Verarta metrics

Content — First Blog Posts (ongoing, 1-2 hours spare time)

- ☐ Write first blog post for amiheines.com (derived from setup work this week):
  - Candidate: “How I Self-Host My Entire Business Infrastructure” or “Enterprise Monitoring for $0”
- ☐ Publish to amiheines.com/blog
- ☐ Share on LinkedIn as a post with key takeaway

End-of-Week Checkpoints

- ☐ 120+ prospects in spreadsheet/CRM
- ☐ 15-20 LinkedIn messages sent, tracking responses
- ☐ Listmonk welcome sequence live and tested
- ☐ Grafana has Server, Docker, and Blockchain dashboards with real data
- ☐ ChainVault and Verarta exposing `/metrics` to Prometheus
- ☐ First blog post published

Week 3 — Monitoring Polish + ChainVault Stripe & OAuth

Dates: _____ to _____ **Focus:** 40% monitoring completion / 40% ChainVault / 20% outreach continuation
Target hours: 20

Monitoring Polish (8 hours)

M7 — Per-App Grafana Dashboards (2-3 hours)

- ☐ Create ChainVault App dashboard:
  - API request rate + error rate
  - Latency p50/p95/p99
  - Active deals gauge
  - Document uploads/day
  - Wire verifications by result
  - Stripe payment outcomes
- ☐ Create Verarta App dashboard:
  - API request rate
  - Artwork uploads/day
  - Chunk upload throughput
  - WebAuthn auth success/failure ratio
  - Pace controller state
- ☐ Provision as JSON dashboard files

M8 — Database Exporters (1 hour)

- ☐ Add to docker-compose.monitoring.yml:
  - `postgres-exporter` (pointing to ChainVault + Verarta PostgreSQL instances)
  - `redis-exporter` (pointing to Redis instances)
  - `elasticsearch-exporter` (pointing to ES clusters)
- ☐ Add scrape targets to Prometheus config
- ☐ Test: all exporters reporting metrics

M9 — Database Grafana Dashboards (2 hours)

- ☐ Create PostgreSQL dashboard: active connections, queries/sec, cache hit ratio, table sizes, slow queries
- ☐ Create Redis dashboard: memory usage, key count, hit/miss ratio, connected clients
- ☐ Create Elasticsearch dashboard: cluster health, shard count, indexing rate, search latency
- ☐ Provision as JSON dashboard files

M10 — Alertmanager (1 hour)

- ☐ Configure Alertmanager with MiaB SMTP (ami@amiheines.com)
- ☐ Write alert rules in Prometheus:
  - Host disk usage > 80%
  - Container restart count > 3 in 5 minutes
  - Any scrape target down for > 2 minutes
  - Node memory usage > 90%
- ☐ Test: trigger a test alert, verify email arrives

M11 — Uptime Kuma (1-2 hours)

- ☐ Deploy Uptime Kuma container
- ☐ Configure endpoint monitors:
  - amiheines.com (HTTPS)
  - chainvault.amiheines.com (HTTPS)
  - verarta.com (HTTPS)
  - pofov.com (HTTPS)

    - lists.amiheines.com (HTTPS)
    - matomo.amiheines.com (HTTPS)
- ☐ Set up public status page at status.amiheines.com
- ☐ Configure email notifications on downtime

M12 — Nginx Configs (30 min)

- ☐ Create nginx config for monitoring.amiheines.com → Grafana (auth-protected)
- ☐ Create nginx config for status.amiheines.com → Uptime Kuma (public)
- ☐ Run certbot for both domains
- ☐ Test: access both URLs, verify SSL

ChainVault Expansion Start (8 hours)

CV1 — Stripe Live Integration (3-4 hours)

- ☐ Add Stripe live API keys to production .env
- ☐ Test webhook endpoint with Stripe CLI (`stripe listen --forward-to`)
- ☐ Create test subscription lifecycle: sign up → trial → payment → invoice → cancel
- ☐ Verify billing_subscriptions and billing_usage tables update correctly
- ☐ Test edge cases: failed payment, card update, plan change
- ☐ Confirm email notifications fire on billing events

CV2 — OAuth SSO Completion (3-4 hours)

- ☐ Configure Google OAuth app (Google Cloud Console):
    - Set authorized redirect URIs
    - Add amiheines.com domain
- ☐ Configure Microsoft Entra ID app:
    - Register app in Azure Portal
    - Set redirect URIs
    - Configure permissions
- ☐ Wire up Auth.js providers in ChainVault Next.js config
- ☐ Test login flow: Google sign-in → account creation → session → dashboard
- ☐ Test login flow: Microsoft sign-in → account creation → session → dashboard
- ☐ Verify SSO users can access all features (deals, documents, wire verification)

Outreach Continuation (4 hours)

- ☐ Send 10-15 new personalized LinkedIn messages
- ☐ Follow up on week 2 messages (anyone who viewed but didn't respond)
- ☐ Post 3-5 LinkedIn insights this week (short, opinion-driven)
- ☐ Comment on 10+ prospect posts (add value, build visibility)
- ☐ Track metrics: messages sent, responses, calls booked
- ☐ If any discovery calls happen: run them, take notes, iterate pitch

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Content (spare time)

- ☐ Write/publish blog post #2 (candidate: "Why Your Sysadmin Can Edit Your Audit Logs")
- ☐ Share on LinkedIn

End-of-Week Checkpoints

- ☐ All monitoring dashboards live (Server, Docker, Blockchain, Apps, DBs)
- ☐ Alertmanager sending email alerts
- ☐ Uptime Kuma public status page at status.amiheines.com
- ☐ monitoring.amiheines.com accessible with auth
- ☐ ChainVault Stripe live and tested end-to-end
- ☐ ChainVault Google + Microsoft SSO working
- ☐ 30-35 total LinkedIn outreach messages sent
- ☐ Blog post #2 published

Week 4 — ChainVault Monitoring + Logging + Alerts

Dates: _____ to _____ **Focus:** 70% ChainVault observability / 20% outreach / 10% content **Target hours:** 20

ChainVault Expansion (14 hours)

CV3 — Prometheus Metrics Enhancement (3-4 hours)

- ☐ Expand chain-service metrics:
  - `chainvault_wire_verifications_total{result}` (match/mismatch/flagged)
  - `chainvault_stripe_payments_total{status}` (succeeded/failed/refunded)
  - `blockchain_transaction_duration_seconds` (histogram)
  - `chainvault_documents_uploaded_total` (counter)
  - `chainvault_deals_total{status}` (gauge: open/closed/cancelled)
- ☐ Add metrics to Next.js API middleware:
  - Per-route latency and error tracking
  - `http_request_size_bytes` / `http_response_size_bytes`
- ☐ Add metrics to hyperion-sync:
  - `hyperion_sync_actions_processed_total`
  - `hyperion_sync_lag_seconds`
  - `hyperion_sync_reconnections_total`
- ☐ Update Prometheus scrape config for new endpoints
- ☐ Test: all new metrics visible in Prometheus

CV4 — Structured Logging (2-3 hours)

- ☐ Install `pino` in chain-service
- ☐ Replace all `console.log` / `console.error` calls in chain-service/src/index.ts with pino:
  - `logger.info()` for normal operations
  - `logger.error()` for failures (include error stack, request context)
  - `logger.warn()` for retries, rate limits
- ☐ Add request ID to every log entry (generate UUID per request)
- ☐ Replace all logging in hyperion-sync/src/sync.ts with pino
- ☐ Configure pino to output JSON (Promtail auto-parses JSON)
- ☐ Test: Grafana Explore → Loki → can query ChainVault logs by level, request ID

CV5 — Health Check Dashboard (2 hours)

- ☐ Expand chain-service `GET /health` to return detailed status:

```json
{
  "status": "healthy|degraded|unhealthy",
  "blockchain": { "connected": true, "head_block": 12345, "producers_active": 4 },
  "database": { "connected": true, "pool_size": 10, "active": 3 },
  "redis": { "connected": true, "memory_mb": 45 },
  "minio": { "connected": true, "buckets": ["documents"] },
  "hyperion": { "connected": true, "last_indexed_block": 12340 }
}
```

- ☐ Add health check metrics to Prometheus (gauge per component: 1=up, 0=down)
- ☐ Create Grafana health panel: traffic-light status for each component

CV6 — ChainVault Grafana Dashboard (3 hours)

- ☐ Create comprehensive ChainVault dashboard with panels:
  - **Overview row:** total deals, total documents, total users, uptime
  - **API row:** request rate, error rate, latency p50/p95/p99, top endpoints
  - **Blockchain row:** head block, transactions/min, finality lag, producer status

  - **Business row:** deals created/day, documents uploaded/day, wire verifications
  - **Payments row:** Stripe payments by status, revenue (if available)
  - **Infrastructure row:** service health, memory, CPU per container
- ☐ Provision as JSON dashboard file
- ☐ Test: all panels render with real data

CV7 — Alert Rules (2 hours)

- ☐ Add ChainVault-specific Prometheus alert rules:
  - `ChainVaultBlockchainStopped` — head block unchanged for > 5 minutes
  - `ChainVaultAPIErrorRateHigh` — error rate > 5% for 5 minutes
  - `ChainVaultStripeWebhookFailed` — any Stripe webhook failure
  - `ChainVaultDiskUsageHigh` — disk > 80%
  - `ChainVaultServiceRestarted` — container restart detected
  - `ChainVaultHyperionLag` — sync lag > 100 blocks
  - `ChainVaultHealthDegraded` — health endpoint returns degraded/unhealthy
- ☐ Test each alert by simulating the condition where possible
- ☐ Verify email delivery via Alertmanager

Outreach (4 hours)

- ☐ Send 10-15 new LinkedIn messages
- ☐ Follow up on prior messages (2nd touch for non-responders from week 2)
- ☐ Post 3-5 LinkedIn insights
- ☐ Comment on 10+ prospect posts
- ☐ Any discovery calls: run, take notes, send follow-up proposals
- ☐ Running total: 45-50 outreach messages sent

Content (2 hours)

- ☐ Write/publish blog post #3
  - Candidate: “Monitoring a Private Blockchain: What Metrics Actually Matter” (from CV3-CV6 work)
- ☐ Share on LinkedIn
- ☐ Set up YouTube channel if not done: name, branding, thumbnail template, channel description

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End-of-Week Checkpoints

- ☐ ChainVault fully observable: metrics, structured logs, health checks, alerts
- ☐ ChainVault Grafana dashboard showing real production data
- ☐ All alert rules tested and email delivery confirmed
- ☐ 45-50 total outreach messages sent, tracking response rate
- ☐ Blog post #3 published
- ☐ YouTube channel set up (even if no videos yet)

Week 5 — ChainVault Finish + Verarta Start

Dates: _____ to _____ **Focus:** 50% ChainVault completion / 30% Verarta start / 20% outreach + content **Target hours:** 20

ChainVault Completion (10 hours)

CV8 — Smart Contract Deployment (2-3 hours)

- ☐ Compile chainvault, auditlog, wirevault contracts (C++ → WASM + ABI)
- ☐ Deploy to production chain:
  - Create contract accounts if not existing
  - Set contract code + ABI via `cleos set contract`
  - Set account permissions
- ☐ Verify deployment via Hyperion API (query contract tables)
- ☐ Test: push a sample transaction, confirm it appears in Hyperion history
- ☐ Document deployment steps in DEPLOYMENT.md

CV9 — Demo Data Seeder (2 hours)

- ☐ Write seeder script (Node.js / TypeScript):
  - Create 2-3 demo organizations
  - Create 5-10 demo users across orgs
  - Create 10-15 demo deals (mix of open, closed, cancelled)
  - Upload 5-10 demo documents (small test files, encrypted)
  - Create 3-5 wire verification records (match, mismatch, flagged)
  - Generate audit trail entries
- ☐ Make script idempotent (safe to re-run)
- ☐ Run on production to populate demo data
- ☐ Verify: data appears correctly in UI and on chain

CV10 — Public Demo Mode (3-4 hours)

- ☐ Create demo login endpoint: `POST /api/auth/demo`
  - Returns session for read-only demo user
  - Demo user has access to all demo org data
  - All write operations return “Demo mode — read only” error
- ☐ Add “Try Demo” button on ChainVault landing page
- ☐ Add “Demo Mode” badge in UI header when demo session active
- ☐ Restrict demo session: no document downloads, no real Stripe, no email sends
- ☐ Auto-expire demo sessions after 30 minutes
- ☐ Test: click demo → explore deals, documents, audit trail, wire verifications → all read-only
- ☐ Add demo link to amiheines.com/portfolio

Verarta Start (6 hours)

V1 — Email Infrastructure (3-4 hours)

- ☐ Configure MiaB SMTP credentials in Verarta backend .env
- ☐ Implement actual email sending in email.ts (replace DEV_MODE bypass):
  - Verification code emails (6-digit code, styled HTML template)
  - Welcome email after verification
  - Artwork share notifications
- ☐ Test SMTP delivery: send to test addresses, check deliverability (SPF, DKIM, DMARC)
- ☐ Handle SMTP errors gracefully (retry once, log failure, don’t block user action)
- ☐ Verify: full registration flow works with real email delivery

V2 — Verarta Prometheus Metrics (2-3 hours)

- ☐ Expand existing `prom-client` setup with Verarta-specific metrics:
  - `verarta_artworks_registered_total` — artwork registration counter
  - `verarta_chunks_uploaded_total` — chunk throughput counter
  - `verarta_chunk_upload_duration_seconds` — histogram
  - `verarta_webauthn_auth_total{result}` — auth success/failure
  - `verarta_quota_usage_ratio{user}` — gauge
  - `verarta_pace_controller_state` — gauge (SLOW=0, BURST=1)
- ☐ Verify Prometheus scrapes new metrics
- ☐ Spot-check metrics in Grafana Explore

Outreach + Content (4 hours)

Outreach (2-3 hours)

- ☐ Send 10-15 new LinkedIn messages
- ☐ Follow up on prior non-responders (3rd touch for week 2, 2nd for week 3)
- ☐ Post 3-5 LinkedIn insights
- ☐ Any discovery calls or proposals
- ☐ Running total: 55-65 outreach messages sent
- ☐ **First revenue check:** any paid engagements closed? Any in pipeline?

Content (1-2 hours)

- ☐ Publish blog post #4 or first YouTube video
  - Candidate video: “Building a Demo Mode So Prospects Can Try Before They Buy” (from CV10 work)
  - Candidate blog: “How I Built a Wire Fraud Prevention System for Real Estate Closings”
- ☐ Share on LinkedIn

End-of-Week Checkpoints

- ☐ ChainVault 100% complete: contracts deployed, demo data, demo mode live
- ☐ ChainVault demo link added to amiheines.com/portfolio
- ☐ Verarta email infrastructure working (real email delivery)
- ☐ Verarta metrics in Prometheus
- ☐ 55-65 total outreach messages, tracking conversion
- ☐ 4 blog posts or 1 video + 3 posts published

Week 6 — Verarta Logging + Monitoring + File Pipeline

Dates: _____ to _____ **Focus:** 80% Verarta expansion / 20% outreach + content **Target hours:** 20

Verarta Expansion (16 hours)

V3 — Structured Logging (3-4 hours)

- ☐ Install `pino` in Verarta backend
- ☐ Replace `console.error()` in all 32 API endpoints with pino structured logs:
  - `logger.info()` for successful operations
  - `logger.error()` with full error context (stack, request body, user)
  - `logger.warn()` for retries, rate limits, validation failures
- ☐ Add request ID middleware (generate UUID per request, include in all logs)
- ☐ Add user context to logs where available (blockchain account, user ID)
- ☐ Configure pino JSON output for Promtail consumption
- ☐ Test: Grafana → Loki → query Verarta logs by endpoint, level, request ID, user

V4 — Verarta Grafana Dashboard (3 hours)

- ☐ Create Verarta dashboard with panels:
  - **Overview row:** total artworks, total users, active uploads, uptime
  - **API row:** request rate, error rate, latency p50/p95/p99, top endpoints
  - **Blockchain row:** head block, Savanna finality status, pace controller state, transactions/min
  - **Upload row:** artworks registered/day, chunks uploaded, upload duration, failed uploads
  - **Auth row:** WebAuthn registration/login success rate, active sessions
  - **Quota row:** per-user quota utilization, users near limit
- ☐ Provision as JSON dashboard file
- ☐ Test: all panels render with real data

V5 — Alert Rules (1-2 hours)

- ☐ Add Verarta-specific Prometheus alert rules:
  - `VerartaBlockchainFinalityLag` — finality lag > 30 seconds
  - `VerartaPaceControllerStuck` — state unchanged for > 10 minutes during activity
  - `VerartaUploadFailureRate` — chunk upload failure > 10% in 5 minutes
  - `VerartaAuthFailureSpike` — WebAuthn failures > 20% in 5 minutes
  - `VerartaAPIErrorRate` — error rate > 5% for 5 minutes
- ☐ Test alerts where possible
- ☐ Verify email delivery

V6 — File Download / Reassembly (4-5 hours)

- ☐ Implement image retrieval pipeline in Verarta backend:
  - Query all chunks for an image from Hyperion history API (ordered by chunk_index)
  - Reassemble chunks in order
  - Decode from base64
  - Serve with correct Content-Type header
- ☐ Implement Redis-cached thumbnails:
  - Generate 300x300 thumbnail on first request (Sharp)
  - Cache in Redis with TTL (24 hours)
  - Serve cached version on subsequent requests
- ☐ Add API endpoint: `GET /api/artworks/:id/image` and `GET /api/artworks/:id/thumbnail`
- ☐ Handle edge cases: incomplete uploads, missing chunks, corrupted data
- ☐ Test: upload artwork → retrieve full image → verify matches original → verify thumbnail

V7 — Public Gallery (4-5 hours)

- ☐ Create public gallery page at verarta.com:
  - Grid of registered artworks with thumbnails
  - Click through to individual artwork page
  - Show provenance chain: registration date, blockchain transaction, artist info
  - Link to blockchain explorer for each artwork (transaction ID)
- ☐ Add artwork visibility controls (artist can mark artwork as public/private)
- ☐ SEO: meta tags, Open Graph, structured data for artworks
- ☐ Add gallery link to amiheines.com/portfolio
- ☐ Test: public gallery renders, images load from chain, provenance links work

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Outreach + Content (4 hours)

Outreach (2-3 hours)

- ☐ Send 10-15 new LinkedIn messages
- ☐ Follow up on pipeline (anyone close to booking a call?)
- ☐ Post 3-5 LinkedIn insights
- ☐ Comment on prospect posts
- ☐ Running total: 65-80 outreach messages

Content (1-2 hours)

- ☐ Publish blog post #5 or YouTube video #2
  - Candidate: “Reassembling Files Stored on a Blockchain — The Art Provenance Pipeline” (from V6 work)
  - Candidate: “Digital Provenance for Physical Art: How Verarta Works Under the Hood”
- ☐ Share on LinkedIn

End-of-Week Checkpoints

- ☐ Verarta has structured logging, queryable in Grafana
- ☐ Verarta Grafana dashboard live with real data
- ☐ Verarta alert rules active and tested
- ☐ File download/reassembly pipeline working (upload → retrieve → matches)
- ☐ Public gallery page live at verarta.com
- ☐ Gallery linked from amiheines.com/portfolio
- ☐ 65-80 outreach messages total

Week 7 — Verarta Multi-Device WebAuthn + Finish

Dates: _____ to _____ **Focus:** 70% Verarta completion / 20% outreach / 10% content **Target hours:** 20

Verarta Expansion (14 hours)

V8 — Multi-Device WebAuthn Signers (6-8 hours)

This is the biggest single feature. Break it into sub-tasks:

Smart Contract Changes (2 hours)

- ☐ Add `devices` table to verarta.core contract:
  - device_id, user account, credential_id_hash, nickname, registered_at, last_used_at, is_active
- ☐ Add `recovery` table:
  - recovery_id, user account, trusted_accounts (vector of names), threshold (e.g., 2), created_at
- ☐ Add actions: `adddevice`, `rmdevice`, `setrecovery`, `execrecovery`
- ☐ Compile and deploy updated contract

Backend (2-3 hours)

- ☐ API: `POST /api/auth/devices/register` — register additional WebAuthn device to existing account
- ☐ API: `GET /api/auth/devices` — list user's registered devices (nickname, last used, active)
- ☐ API: `DELETE /api/auth/devices/:id` — revoke a device
- ☐ API: `POST /api/auth/recovery/setup` — configure trusted contacts (2-of-N)
- ☐ API: `POST /api/auth/recovery/initiate` — start recovery process
- ☐ API: `POST /api/auth/recovery/approve` — trusted contact approves recovery
- ☐ Update login flow: try all registered credentials, update last_used_at on success

Frontend (2-3 hours)

- ☐ Settings page: "My Devices" section
  - List all registered devices with nickname, last used date
  - "Add Device" button → WebAuthn registration flow → nickname prompt
  - "Revoke" button per device (with confirmation)
- ☐ Settings page: "Account Recovery" section
  - Configure trusted contacts (search by username/account)
  - Set threshold (minimum approvals needed)
  - Show current recovery config
- ☐ Recovery flow: "Lost access?" → enter username → trusted contacts notified → approvals → new device registration
- ☐ Test: register 2 devices, login with each, revoke one, verify can't login with revoked

V9 — Demo Data Seeder (2 hours)

- ☐ Write seeder script:
  - Create 3-5 demo artist accounts
  - Register 10-15 demo artworks (use sample images, varied sizes)
  - Upload artwork files to chain (chunked)
  - Create provenance records
  - Set some artworks as public, some private
- ☐ Make script idempotent
- ☐ Run on production
- ☐ Verify: gallery shows demo artworks, provenance chains visible

V10 — Public Demo Mode (3-4 hours)

- ☐ Create demo login: `POST /api/auth/demo`
  - Read-only session for demo user
  - Access to demo artworks, provenance, blockchain explorer
  - All write operations blocked with "Demo mode" message
- ☐ Add "Try Demo" button on Verarta landing page
- ☐ Add "Demo Mode" badge in UI header
- ☐ Restrict: no file uploads, no real WebAuthn registration, no settings changes
- ☐ Auto-expire after 30 minutes
- ☐ Add demo link to amiheines.com/portfolio
- ☐ Test: full exploration flow in demo mode

Outreach (4 hours)

- ☐ Send 10-15 new LinkedIn messages
- ☐ Follow up on pipeline — prioritize warm leads
- ☐ Post 3-5 LinkedIn insights
- ☐ Comment on prospect posts
- ☐ Running total: 75-95 outreach messages
- ☐ **Revenue check:** How many discovery calls? Any closed deals? Pipeline value?
- ☐ If no calls yet: review message templates, try different angles, consider warm intro requests

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Content (2 hours)

- ☐ Publish blog post #6 or YouTube video
  - Candidate: "Multi-Device WebAuthn for Blockchain Accounts: A Recovery Strategy" (from V8)
  - Candidate video: first "Build With Me" recording (record while working on V8)
- ☐ Share on LinkedIn

End-of-Week Checkpoints

- ☐ Verarta multi-device WebAuthn working (add, list, revoke devices)
- ☐ Account recovery configured and tested (2-of-N trusted contacts)
- ☐ Verarta demo data populated, gallery shows real artworks
- ☐ Verarta demo mode live with "Try Demo" button
- ☐ Verarta demo linked from amiheines.com/portfolio
- ☐ **Both ChainVault and Verarta now fully expanded, monitored, and demo-ready**
- ☐ 75-95 outreach messages total

Week 8 — Parking Tickets: Smart Contract + Backend Foundation

Dates: _____ to _____ **Focus:** 70% Parking Tickets new build / 20% outreach / 10% content **Target hours:** 20

Parking Tickets — New Build Start (14 hours)

P1 — Smart Contract: parking.core (6-8 hours)

Contract Code (4-5 hours)

- ☐ Create project structure: `parking/blockchain/contracts/parking.core/`
- ☐ Write `parking.core.hpp` — tables and action declarations:
  - `cities` table (city_id, admin_account, city_name, currency, timezone, deadline_days, created_at, is_active)
  - `priceopts` table (option_id, city_id, label, amount, is_active)
  - `employees` table (employee_id, city_id, account, display_name, badge_number, is_active, added_at)
  - `tickets` table (ticket_id, city_id, employee_id, license_plate, location, option_id, amount, image_count, status, issued_at, deadline_at, paid_at, payment_ref, notes)
  - `tktimages` table (image_id, ticket_id, uploader, filename, mime_type, file_size, file_hash, total_chunks, uploaded_chunks, upload_complete)
  - `tktchunks` table (chunk_id, image_id, uploader, chunk_index, chunk_data, chunk_size)
  - Secondary indices: by_city, by_employee, by_plate, by_status, by_ticket, by_image
- ☐ Write `parking.core.cpp` — action implementations:
  - `createcity`, `updatecity` (admin auth)
  - `addpriceopt`, `rmpriceopt` (admin auth)
  - `addemployee`, `rmemployee` (admin auth)
  - `issueticket` (employee auth, validates city/employee/priceopt)
  - `addimage`, `uploadchunk`, `completeimg` (employee auth, chunked upload)
  - `payticket` (backend service auth)
  - `cancelticket` (admin auth)
  - `expireticket` (backend service auth)
- ☐ Write `CMakeLists.txt`

Compile + Deploy (2-3 hours)

- ☐ Compile with Antelope CDT (eosio-cpp)
- ☐ Create `parkingcore` account on Verarta blockchain
- ☐ Deploy WASM + ABI to `parkingcore`
- ☐ Test all actions via cleos:
  - Create a test city
  - Add price options
  - Add an employee
  - Issue a ticket
  - Upload image chunks, complete
  - Pay, cancel, expire tickets
- ☐ Verify via Hyperion: all actions and table deltas indexed

P2 — Backend Foundation (4-5 hours)

- ☐ Initialize Astro 5 SSR project: `parking/backend/`
- ☐ Install dependencies: @astrojs/node, pg, redis, zod, jsonwebtoken, bcrypt, nodemailer, prom-client
- ☐ Create PostgreSQL database `parking`
- ☐ Write and run migration `001_initial_schema.sql`:
  - admins, employees, cities, sessions, tickets, ticket_images, image_chunk_uploads, payment_logs, vehicle_registry_cache
- ☐ Create core libs:
  - `db.ts` — PostgreSQL connection pool
  - `redis.ts` — Redis client
  - `auth.ts` — JWT + bcrypt password hashing + session management

    - `antelope.ts` — Antelope/Spring chain client (reuse pattern from Verarta)
    - `email.ts` — Nodemailer SMTP via MiaB
- ☐ Create middleware: `auth.ts` — JWT verification, user type extraction (admin/employee)
- ☐ Create Astro config: SSR mode, Node adapter, port 4322
- ☐ Create PM2 ecosystem config
- ☐ Test: backend starts, connects to DB + Redis + chain, health endpoint responds

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Outreach (4 hours)

- ☐ Send 10-15 new LinkedIn messages
- ☐ Follow up on pipeline
- ☐ Post 3-5 LinkedIn insights
- ☐ Running total: 85-110 outreach messages
- ☐ **Revenue milestone check:** Should be approaching first paid engagement by now
- ☐ If calls are happening: refine proposal template based on feedback

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Content (2 hours)

- ☐ Publish blog post #7 or YouTube video
    - Candidate: "Writing an Antelope Smart Contract for Parking Tickets — From Scratch" (from P1 work — great "Build With Me" video)
- ☐ Record development session if doing video (screen capture while coding P1)

End-of-Week Checkpoints

- ☐ parking.core smart contract compiled, deployed, and tested on Verarta chain
- ☐ All contract actions verified via cleos and Hyperion
- ☐ Parking backend running: DB, Redis, chain connection, auth, health endpoint
- ☐ 85-110 outreach messages total
- ☐ Revenue pipeline review done

Week 9 — Parking Tickets: Admin Flow + Ticketing

Dates: _____ to _____ **Focus:** 80% Parking backend features / 20% outreach + content **Target hours:** 20

Parking Backend — Admin + Ticketing (16 hours)

P3 — Admin Auth + City Setup (5-6 hours)

Authentication (2-3 hours)

- ☐ `POST /api/auth/admin/register` — email + password registration
- ☐ `POST /api/auth/verify-email` — 6-digit verification code
- ☐ `POST /api/auth/admin/login` — login, return JWT
- ☐ `POST /api/auth/logout` — terminate session
- ☐ `GET /api/auth/session` — get current session info
- ☐ Send verification emails via MiaB SMTP
- ☐ Test: full registration → verification → login → session flow

City Management (1-2 hours)

- ☐ `POST /api/cities` — create city:
  - Create blockchain account for admin
  - Push `createcity` action to chain
  - Insert into PostgreSQL
- ☐ `PUT /api/cities/:id` — update city settings (chain + DB)
- ☐ `GET /api/cities/:id` — get city details
- ☐ `GET /api/cities/:id/stats` — dashboard statistics query

Price Options (1 hour)

- ☐ `GET /api/cities/:id/prices` — list price options
- ☐ `POST /api/cities/:id/prices` — add price option (chain + DB)
- ☐ `PUT /api/prices/:id` — update price option
- ☐ `DELETE /api/prices/:id` — deactivate (chain + DB)

Employee Management (1 hour)

- ☐ `GET /api/cities/:id/employees` — list employees
- ☐ `POST /api/cities/:id/employees` — invite employee:
  - Create blockchain account
  - Push `addemployee` action
  - Send invite email with temp credentials
- ☐ `PUT /api/employees/:id` — update employee
- ☐ `DELETE /api/employees/:id` — deactivate (chain + DB)
- ☐ `POST /api/auth/employee/login` — employee login

P4 — Ticket Issuance (4-5 hours)

- ☐ `POST /api/tickets` — issue a new ticket:
  - Validate: employee auth, city, price option exists
  - Look up vehicle owner from registry (see below)
  - Push `issueticket` action to chain
  - Insert into PostgreSQL (with deadline_at = now + deadline_days)
  - Return ticket ID + details
- ☐ `GET /api/tickets` — list tickets with filters (city, status, plate, date range, employee)
- ☐ `GET /api/tickets/:id` — ticket detail with images
- ☐ `PUT /api/tickets/:id/cancel` — admin cancel (chain + DB)

Vehicle Registry Mock (1 hour)

- ☐ `GET /api/vehicles/:plate` — lookup vehicle owner
- ☐ Implement mock mode: return fake owner data based on plate prefix
- ☐ Implement cache: check `vehicle_registry_cache` first (24h TTL)
- ☐ Interface ready for real registry API integration later

P5 — On-Chain Image Upload (5-6 hours)

- ☐ `POST /api/tickets/:id/images/init` — initialize image upload:
  - Accept full file (multipart, max 10MB)
  - Validate: JPEG/PNG/WebP (magic bytes, not just extension)
  - Save to temp disk, compute SHA256, calculate chunk count
  - Push `addimage` action to chain
  - Insert into ticket_images + create tracking rows
  - Return image_id, total_chunks
- ☐ `POST /api/tickets/:id/images/:imageId/chunk` — upload one chunk:
  - Read 256KB from temp file at correct offset
  - Base64 encode
  - Push `uploadchunk` action to chain
  - Update image_chunk_uploads tracking
  - Return chunk_index, chunks_remaining
- ☐ `POST /api/tickets/:id/images/:imageId/complete` — complete upload:
  - Verify all chunks uploaded
  - Push `completeimg` action to chain
  - Delete temp file
  - Update ticket image_count on chain
- ☐ `GET /api/tickets/:id/images` — list images for a ticket
- ☐ `GET /api/tickets/:id/images/:imageId` — reconstruct and serve image:
  - Query chunks from Hyperion by image_id
  - Reassemble in order by chunk_index
  - Decode base64
  - Serve with correct Content-Type
- ☐ Thumbnail generation: Sharp, 300x300, Redis cached
- ☐ Test: upload JPEG → all chunks on chain → retrieve → matches original

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Outreach + Content (4 hours)

Outreach (3 hours)

- ☐ Send 10-15 new LinkedIn messages
- ☐ Follow up on warm leads
- ☐ Post 3-5 LinkedIn insights
- ☐ Running total: 95-125 outreach messages
- ☐ Focus on closing pipeline: anyone who had a discovery call — send follow-up proposal

Content (1 hour)

- ☐ Publish blog post #8
  - Candidate: “Storing Evidence Photos on a Blockchain: Chunked Upload Deep Dive” (from P5)
- ☐ Share on LinkedIn

End-of-Week Checkpoints

- ☐ Parking admin flow complete: registration, city setup, prices, employees
- ☐ Ticket issuance working: create, list, filter, detail, cancel
- ☐ On-chain image upload pipeline working end-to-end

- ☐ Vehicle registry mock responding
- ☐ 95-125 outreach messages total

Week 10 — Parking Tickets: Payments + Expiration + Email

Dates: _____ to _____ **Focus:** 80% Parking backend completion / 20% outreach + content **Target hours:** 20

Parking Backend Completion (16 hours)

P6 — Stripe Payment Integration (4-5 hours)

Payment Flow (3-4 hours)

- ☐ Install Stripe SDK in parking backend
- ☐ Create `stripe.ts` lib: client initialization, helper functions
- ☐ Generate secure payment tokens (UUID v4 + HMAC) for each ticket
- ☐ `GET /api/public/tickets/:token` — public ticket view (no auth):
  - Validate token
  - Return: ticket details, violation, amount, deadline, images (thumbnails)
  - Don't expose internal IDs or employee info
- ☐ `POST /api/public/tickets/:token/pay` — create Stripe PaymentIntent:
  - Validate ticket is open and not expired
  - Create PaymentIntent with ticket amount + currency
  - Return client_secret for Stripe Elements
- ☐ `POST /api/webhooks/stripe` — handle Stripe webhook:
  - Verify webhook signature
  - On `payment_intent.succeeded`:
    - Push `payticket` action to chain (with Stripe payment ID)
    - Update ticket status to "paid" in PostgreSQL
    - Send payment confirmation email to car owner
    - Log in payment_logs table
  - On `payment_intent.payment_failed`:
    - Log failure, don't change ticket status

Testing (1 hour)

- ☐ Test with Stripe test keys: full flow from link → view → pay → confirmed
- ☐ Test edge cases: expired ticket payment attempt, double payment, invalid token
- ☐ Verify: blockchain has `payticket` action, DB has correct status, email sent

P7 — Deadline Expiration (2 hours)

- ☐ Install node-cron in parking backend
- ☐ Create `cron.ts`:
  - Daily job at 02:00 local time
  - Query: `SELECT * FROM tickets WHERE status = 'open' AND deadline_at < NOW()`
  - For each expired ticket:
    - Push `expireticket` action to chain
    - Update status to "expired" in PostgreSQL
    - Send final notice email to car owner
    - Notify city admin (optional summary email)
- ☐ Add cron initialization to Astro server startup
- ☐ Test: create ticket with past deadline → run cron manually → verify status change + email
- ☐ Log cron runs and results (pino)

P11 — Email Templates (2-3 hours)

- ☐ Create HTML email templates:
  - **Ticket Issued:** ticket details, violation, amount, deadline, photo thumbnails, payment link button
  - **Payment Confirmation:** receipt with ticket summary, amount paid, transaction reference
  - **Deadline Reminder** (7 days before): reminder with payment link, countdown to deadline
  - **Ticket Expired:** final notice, escalation warning

    - **Admin: Payment Received:** notification with ticket + payment details
    - **Employee Invite:** login credentials + getting started instructions
- ☐ Make templates responsive (mobile email clients)
- ☐ Add email sending to:
    - Ticket issuance flow (P4)
    - Payment success (P6)
    - Expiration cron (P7)
- ☐ Create deadline reminder cron: 7 days before deadline, send reminder for open tickets
- ☐ Test: verify all templates render correctly, links work, images display

Parking Backend — Wire Everything Together (4-5 hours)

- ☐ End-to-end integration test of full flow:
    1. Admin registers → verifies email → logs in
    2. Admin creates city → adds price options → adds employee
    3. Employee logs in → issues ticket (with location, plate, violation, photo)
    4. Photo uploads to chain (chunked) → progress tracked
    5. Car owner email received with payment link
    6. Car owner clicks link → views ticket + photos → pays via Stripe
    7. Payment confirmed → ticket marked paid on chain + DB → confirmation email
    8. Separate ticket: not paid → deadline passes → cron expires → final notice email
- ☐ Fix any integration bugs discovered
- ☐ Add error handling for edge cases found during testing
- ☐ Verify all blockchain transactions are indexed in Hyperion

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Outreach + Content (4 hours)

Outreach (3 hours)

- ☐ Send 10-15 new LinkedIn messages
- ☐ Nurture pipeline: follow up on proposals, answer questions
- ☐ Post 3-5 LinkedIn insights
- ☐ Running total: 105-140 outreach messages
- ☐ **Revenue milestone:** Target first $500-$1,500 paid engagement by end of week 10

Content (1 hour)

- ☐ Publish blog post #9 or YouTube video
    - Candidate: “Stripe + Blockchain: Marking a Parking Ticket Paid On-Chain” (from P6)
- ☐ Share on LinkedIn

End-of-Week Checkpoints

- ☐ Parking backend 100% feature-complete:
    - Admin flow, employee flow, ticketing, images, payments, expiration, emails
- ☐ Full end-to-end flow tested and working
- ☐ All blockchain transactions verified in Hyperion
- ☐ 105-140 outreach messages total
- ☐ Revenue pipeline review: deals closed, proposals pending, calls scheduled

Week 11 — Parking Tickets: Frontend — Admin Dashboard

Dates: _____ to _____ **Focus:** 80% Parking frontend / 20% outreach + content **Target hours:** 20

Parking Frontend — Admin (16 hours)

Initialize Frontend Project (2 hours)

- ☐ Create `parking/frontend/` Next.js 16 project:
  - App Router, React 19, TypeScript
  - Tailwind CSS 4
  - Zustand 5 (state management)
  - TanStack Query 5 (data fetching)
- ☐ Set up API client module (`lib/api/`) with typed fetch helpers
- ☐ Set up auth store (Zustand): login state, JWT token, user info
- ☐ Create base layout with header, navigation, footer
- ☐ Configure next.config.ts: API proxy to parking backend

P8 — Admin Frontend (8-10 hours)

Auth Pages (2 hours)

- ☐ `/auth/admin/register` — registration form (email, password, confirm password)
- ☐ `/auth/admin/login` — login form
- ☐ Email verification page (enter 6-digit code)
- ☐ Auth redirect logic: unauthenticated → login, authenticated → dashboard
- ☐ Login persistence (JWT in httpOnly cookie or localStorage)

City Setup Wizard (2 hours)

- ☐ `/admin/setup` — first-time setup wizard (shown when admin has no city):
  - Step 1: City name, currency, timezone
  - Step 2: Add first price options (violation types + amounts)
  - Step 3: Invite first employee (optional)
  - Step 4: Confirmation + "Go to Dashboard"
- ☐ Wizard state in Zustand, submit all at once
- ☐ Redirect to dashboard after completion

Admin Dashboard (2-3 hours)

- ☐ `/admin` — dashboard with stats cards:
  - Total tickets (open / paid / expired / cancelled)
  - Revenue this month / this week / today
  - Tickets issued today
  - Payment conversion rate
  - Upcoming expirations (next 7 days)
- ☐ Recent tickets list (last 10, clickable)
- ☐ Quick stats chart: tickets over time (last 30 days)

Ticket Management (2-3 hours)

- ☐ `/admin/tickets` — filterable ticket list:
  - Filters: status, date range, license plate, employee, violation type
  - Sortable columns: date, amount, status
  - Pagination
  - Click row → ticket detail
- ☐ `/admin/tickets/[id]` — ticket detail view:
  - Full ticket info: violation, amount, location (map embed), plate, employee, dates
  - Image gallery (thumbnails, click to expand)

    - Blockchain transaction link
    - Status badge (open/paid/expired/cancelled)
    - Cancel button (admin only, with confirmation modal)
    - Payment info (if paid): Stripe ref, date, amount

Employee & Price Management (2 hours)

- ☐ `/admin/employees` — employee list:
    - Table: name, badge, email, status, tickets issued
    - Add employee form (modal: name, email, badge number)
    - Deactivate toggle
- ☐ `/admin/prices` — price options list:
    - Table: label, amount, status
    - Add price option form (modal: label, amount)
    - Deactivate toggle
    - Inline edit for label and amount

Admin Layout & Navigation (1-2 hours)

- ☐ Admin sidebar navigation:
    - Dashboard, Tickets, Employees, Prices, Settings
    - City name + currency in sidebar header
    - Logout at bottom
- ☐ Responsive: sidebar collapses to hamburger on mobile
- ☐ Breadcrumbs on detail pages
- ☐ Loading states (skeletons) on all data-dependent pages

Outreach + Content (4 hours)

Outreach (3 hours)

- ☐ Send 10-15 new LinkedIn messages
- ☐ Nurture pipeline: follow up, answer questions
- ☐ Post 3-5 LinkedIn insights
- ☐ Running total: 115-155 outreach messages
- ☐ Close pending proposals — push for commitment

Content (1 hour)

- ☐ Publish blog post #10 or YouTube video
    - Candidate video: “Building a Mobile-First Ticket Issuing App with Next.js and Camera API” (preview of P9 coming next week)
- ☐ Share on LinkedIn

End-of-Week Checkpoints

- ☐ Parking admin frontend fully functional:
    - Registration, login, city setup wizard
    - Dashboard with real stats
    - Ticket list with filters + detail view
    - Employee and price management
- ☐ All admin pages connected to backend API
- ☐ Responsive layout (sidebar, mobile)
- ☐ 115-155 outreach messages total

Week 12 — Parking Tickets: Employee Mobile UI + Public Payment + Deploy

Dates: _____ to _____ **Focus:** 80% Parking frontend completion + deployment / 20% outreach + content **Target hours:** 20

Parking Frontend — Employee + Public + Deploy (16 hours)

P9 — Employee Mobile Frontend (6-8 hours)

Employee Auth (1 hour)

- ☐ `/auth/employee/login` — employee login form
- ☐ Employee session management (same JWT pattern, different user_type)
- ☐ First login: password change prompt

New Ticket Form — Mobile-First (3-4 hours)

- ☐ `/employee/ticket/new` — the core employee workflow:
  - **Camera:** `navigator.mediaDevices.getUserMedia()` integration
    - Take photo button → opens camera
    - Preview captured image, retake option
    - Support multiple photos per ticket
  - **GPS:** `navigator.geolocation.getCurrentPosition()`
    - Auto-capture lat/lng on form open
    - Reverse geocode to human-readable address (free API: Nominatim)
    - Show location on small map (Leaflet)
    - Manual address override option
  - **License plate input:** large text field, uppercase auto-format
  - **Violation type:** radio buttons from city's price options (amount shown per option)
  - **Notes:** optional textarea
  - **Submit button:** prominent, shows upload progress
- ☐ Upload progress UI:
  - Overall progress bar
  - Per-image chunk progress (X of Y chunks uploaded)
  - "Uploading to blockchain…" status message
  - Success confirmation with ticket number

Employee Ticket List (1-2 hours)

- ☐ `/employee` — home page with quick actions: "Issue Ticket" button (prominent)
- ☐ `/employee/tickets` — my issued tickets list (today / this week / all)
- ☐ `/employee/tickets/[id]` — ticket detail (same as admin but employee context)

Employee Layout (1 hour)

- ☐ Mobile-first layout: bottom navigation bar (Home, New Ticket, My Tickets)
- ☐ Header: employee name, badge number, city name
- ☐ Offline handling: show clear message if no network (tickets require connectivity)

P10 — Public Payment Page (3-4 hours)

- ☐ `/pay/[token]` — public-facing payment page (no auth required):
  - Ticket details: violation type, amount, location + map, date, deadline countdown
  - Photo gallery: ticket evidence images (thumbnails from chain)
  - Blockchain verification link: "View on blockchain" → Verarta explorer
  - **Stripe Elements** payment form:
    - Card number, expiry, CVC

        - Pay button with amount
        - Loading state during payment processing
    - Success page: “Payment received” + receipt summary + blockchain transaction link
    - Failure page: “Payment failed” + retry button
- ☐ Already-paid state: “This ticket has been paid” with receipt
- ☐ Expired state: “This ticket has expired” with info
- ☐ Mobile responsive (car owners will open this on their phone from email)

P15 — Demo Data Seeder (2-3 hours)

- ☐ Write seeder script:
    - Create demo city (“Demo City”, USD, America/New_York, 90 days)
    - Create 3 price options (No Parking Zone $75, Expired Meter $50, Fire Hydrant $150)
    - Create 2 demo employees (with badge numbers)
    - Create 15-20 demo tickets:
        - 5 open (various violations, recent dates)
        - 5 paid (with mock payment refs, paid dates)
        - 3 expired (past deadline)
        - 2 cancelled
    - Upload 1-2 demo images per ticket (small JPEG test files, on-chain)
- ☐ Make idempotent
- ☐ Run on production

P16 — Deployment (2-3 hours)

- ☐ Create nginx config for parking.amiheines.com:
    - Frontend (Next.js): proxy to port 3001
    - Backend API: proxy to port 4322 at /api/
    - Stripe webhook: proxy to port 4322 at /api/webhooks/stripe
    - Rate limiting on auth + API endpoints
    - SSL via certbot
- ☐ Create PM2 ecosystem config for parking-backend
- ☐ Create deploy script: build → rsync → restart services
- ☐ Production .env file from template
- ☐ Deploy to server
- ☐ DNS: A record for parking.amiheines.com
- ☐ Test: access parking.amiheines.com, full flow works

Outreach + Content (4 hours)

Outreach (3 hours)

- ☐ Send 10-15 new LinkedIn messages
- ☐ Nurture pipeline, close deals
- ☐ Post 3-5 LinkedIn insights
- ☐ Running total: 125-170 outreach messages
- ☐ **Revenue check:** Should have $500-$2,000 revenue by now or strong pipeline

Content (1 hour)

- ☐ Publish blog post #11 or YouTube video
    - Candidate: “Building a Mobile-First Ticket Issuing App with Next.js and Camera API”
- ☐ Share on LinkedIn

End-of-Week Checkpoints

- ☐ Parking employee mobile UI complete: camera, GPS, ticket form, upload progress
- ☐ Public payment page working: view ticket → pay with Stripe → confirmation
- ☐ Demo data populated: city, employees, 20 tickets with images
- ☐ Parking system deployed to parking.amiheines.com
- ☐ Full end-to-end: admin setup → employee issues ticket → owner pays → all on chain
- ☐ 125-170 outreach messages total

Week 13 — Parking Monitoring + POFOV Lead Gen Start

Dates: _____ to _____ **Focus:** 40% Parking monitoring / 40% POFOV lead gen / 20% outreach + content **Target hours:** 20

Parking Monitoring (8 hours)

P12 — Prometheus Metrics (2-3 hours)

- ☐ Install `prom-client` in parking Astro backend
- ☐ Add metrics:
  - `parking_tickets_issued_total{city, violation_type}`
  - `parking_tickets_paid_total{city}`
  - `parking_tickets_expired_total{city}`
  - `parking_revenue_total_cents{city, currency}`
  - `parking_payment_conversion_rate{city}`
  - `parking_image_upload_duration_seconds`
  - Common: `http_requests_total`, `http_request_duration_seconds`, `app_errors_total`
- ☐ Expose `GET /metrics` endpoint
- ☐ Add Parking scrape target to Prometheus config
- ☐ Test: metrics visible in Prometheus

P13 — Grafana Dashboard (3 hours)

- ☐ Create Parking Tickets dashboard:
  - **Overview row:** total tickets, revenue, active cities, active employees
  - **Tickets row:** tickets issued/day, by violation type breakdown, by city
  - **Revenue row:** payments/day, payment conversion rate, revenue cumulative
  - **Operations row:** image uploads, upload duration, expired tickets/day
  - **API row:** request rate, error rate, latency p50/p95/p99
- ☐ Provision as JSON dashboard file
- ☐ Test: all panels render with demo data

P14 — Alert Rules (1-2 hours)

- ☐ Add Parking-specific alert rules:
  - `ParkingPaymentFailure` — Stripe webhook returns failure
  - `ParkingExpirationCronMissed` — cron didn't run in last 26 hours
  - `ParkingAPIErrorRate` — error rate > 5%
  - `ParkingBlockchainLag` — chain transaction taking > 30 seconds
  - `ParkingImageUploadStuck` — incomplete upload older than 2 hours
- ☐ Test alerts
- ☐ Add parking.amiheines.com to Uptime Kuma

POFOV Lead Gen Start (8 hours)

PF1 — Email Capture Infrastructure (3-4 hours)

- ☐ Create POFOV-specific list in Listmonk
- ☐ Build email signup component for pofov.com:
  - Modal popup (triggered on scroll 50% or after 30 seconds, once per session)
  - Inline form at bottom of every coin page
  - "Get the full report" CTA on coin detail pages
- ☐ Integrate Listmonk API: subscribe endpoint, double opt-in
- ☐ Store subscriber state in cookie (suppress modal for existing subscribers)
- ☐ Test: sign up on pofov.com → appears in Listmonk POFOV list → double opt-in email → confirmed

PF2 — Content Gating System (4-5 hours)

- ☐ Build gating middleware in Astro:
  - Check for `pofov_subscriber` cookie on gated routes
  - If no cookie: show content preview + email wall overlay
  - After email signup: set cookie, Listmonk API confirms subscription, unlock content
  - Cookie expires after 30 days (re-prompt for re-engagement)
- ☐ Implement gated states for:
  - Full AI report (all 5 questions, all 3 providers) → gated
  - Comparison tool (when built) → gated after first comparison
  - CSV export of any data → gated
- ☐ Free state: basic coin page (about, single AI Q&A, stats) always free
- ☐ Test: visit coin page free → click "full report" → email wall → sign up → content unlocked
- ☐ Track gating events in Matomo (impressions, signups, unlock rate)

Outreach + Content (4 hours)

Outreach (3 hours)

- ☐ Send 10-15 new LinkedIn messages
- ☐ Pipeline management: follow up, close, propose
- ☐ Post 3-5 LinkedIn insights
- ☐ Running total: 135-185 outreach messages
- ☐ **Portfolio update:** Add parking.amiheines.com to amiheines.com/portfolio with screenshots and demo link

Content (1 hour)

- ☐ Publish blog post #12 or YouTube video
  - Candidate: "Grafana Dashboard for a Municipal Parking System — Real-Time Revenue Tracking" (from P13)
  - Candidate: "How I Built 3 Blockchain Apps in 14 Weeks Using Claude Code" (milestone recap)
- ☐ Share on LinkedIn

End-of-Week Checkpoints

- ☐ Parking system fully monitored: metrics, dashboard, alerts, uptime
- ☐ **All 3 original projects (ChainVault, Verarta, Parking) complete and monitored**
- ☐ POFOV email capture live: modals, inline forms, Listmonk integration
- ☐ Content gating system working on pofov.com
- ☐ amiheines.com/portfolio updated with all 3 app links
- ☐ 135-185 outreach messages total

Week 14 — POFOV: Comparison Tool + PDF Reports + Enterprise Pages

Dates: _____ to _____ **Focus:** 80% POFOV expansion / 20% outreach + content **Target hours:** 20

POFOV Expansion (16 hours)

PF3 — Comparison Tool (6-8 hours)

- ☐ Create `/compare/[slug1]-vs-[slug2]` dynamic pages:
  - Pull data from both coins via scrapecmc API
  - Side-by-side comparison table:
    - Basic: rank, price, market cap, volume
    - Technical: consensus mechanism, smart contract support, transaction speed, storage capability
    - AI analysis: key differences summary (derived from AI Q&A answers)
    - Use cases: which is better for what
  - Verdict section: "Choose X if… Choose Y if…"
- ☐ Auto-generate top 20 enterprise comparison pairs:
  - Hyperledger Fabric vs Antelope
  - Hyperledger Fabric vs R3 Corda
  - Antelope vs ConsenSys Quorum
  - Hyperledger Fabric vs ConsenSys Quorum
  - (etc. — all permutations of 7 enterprise chains)
- ☐ Gating: summary table free, full comparison (AI analysis + verdict) gated behind email
- ☐ SEO: title tags, meta descriptions, structured data for comparisons
- ☐ Internal linking: from each coin page → "Compare with…" dropdown
- ☐ Test: visit comparison page, free section visible, gated section shows email wall, signup unlocks
- ☐ Matomo tracking: comparison page views, which pairs most popular

PF4 — PDF Report Generator (4-5 hours)

- ☐ Install PDF generation library (puppeteer or @react-pdf/renderer)
- ☐ Create PDF template with POFOV branding:
  - Coin name, symbol, logo
  - All 5 AI questions with answers from all 3 providers
  - Deep scrape data: about, whitepaper summary, technical details
  - Basic stats: rank, market cap, volume
  - Footer: "Generated by POFOV — pofov.com | Built by Ami Heines — amiheines.com"
- ☐ API endpoint: `GET /api/report/:slug` (requires authenticated subscriber)
- ☐ "Download Full Report (PDF)" button on every coin page (gated)
- ☐ Cache generated PDFs (Redis, 24h TTL — regenerate on rescrape)
- ☐ Test: download PDF, verify formatting, verify all data present

PF6 — Enterprise Landing Pages (3-4 hours)

- ☐ Create enhanced `/enterprise/[slug]` pages for all 7 enterprise blockchains:
  - Detailed description (richer than standard coin page)
  - Enterprise use cases (industry-specific)
  - Technical specs table: consensus, TPS, storage, smart contracts, privacy
  - "Who's using it" section (known enterprise adopters)
  - Consulting CTA: "Need help deploying [Platform]? → Book a free 30-min discovery call" → amiheines.com/contact
  - "Get the Enterprise Blockchain Buyer's Guide" → email gated PDF
- ☐ Link enterprise pages from main POFOV navigation
- ☐ SEO optimization: "hyperledger fabric consulting", "antelope blockchain development", etc.
- ☐ Test: all 7 pages render, CTAs link to amiheines.com correctly

PF7 — Cross-Promotion Banners (2-3 hours)

- ☐ Top banner on all POFOV pages:
  - Subtle, dismissible: "Built by Ami Heines — Blockchain Infrastructure Consultant → Learn more"
  - Persistent cookie on dismiss (don't re-show for 7 days)
- ☐ Footer on all pages:
  - "POFOV is a project by Ami Heines. I help companies build tamper-proof infrastructure. → ami-heines.com"
- ☐ Contextual CTAs on enterprise pages:
  - "Not sure which platform is right? → $500 Architecture Review"
- ☐ Comparison tool results CTA:
  - "Need help deciding? → Book a free 30-min discovery call"
- ☐ A/B test banner variants via Matomo (if time permits)
- ☐ Test: banners appear, links work, dismiss persists

Outreach + Content (4 hours)

Outreach (3 hours)

- ☐ Send 10-15 new LinkedIn messages
- ☐ Pipeline management
- ☐ Post 3-5 LinkedIn insights
- ☐ Running total: 145-200 outreach messages
- ☐ Start sharing POFOV comparison pages on LinkedIn: "I compared [X] vs [Y] — here's what I found"

Content (1 hour)

- ☐ Publish blog post #13
  - Candidate: "I Compared 1,000 Blockchain Projects with AI — Here Are the 7 That Work for Enterprise" (uses POFOV data)
  - Candidate: "How to Choose a Private Blockchain Platform: A Framework Using Real Data"
- ☐ Share on LinkedIn + link to POFOV comparison tool

End-of-Week Checkpoints

- ☐ POFOV comparison tool live with 20+ enterprise comparison pages
- ☐ PDF report generation working (gated, branded, cached)
- ☐ 7 enterprise landing pages with consulting CTAs
- ☐ Cross-promotion banners linking to amiheines.com on all POFOV pages
- ☐ 145-200 outreach messages total
- ☐ Starting to see POFOV email signups

Week 15 — POFOV Finish + Email Sequences + Scraper Upgrades

Dates: _____ to _____ **Focus:** 80% POFOV completion / 20% outreach + content **Target hours:** 20

POFOV Completion (16 hours)

PF5 — Capability Matrix Page (4-5 hours)

- ☐ Create `/matrix` page — interactive filterable blockchain comparison table:
  - Columns: Platform, Consensus, Smart Contracts (yes/no), On-Chain Storage (yes/no), Privacy Features, TPS Claimed, Enterprise Ready, Category
  - Data sources: AI scrape answers (parsed into structured fields) + manual curation for 7 enterprise chains
  - Filters: by capability (checkboxes), by category, by enterprise-ready
  - Sort: by any column
  - Search: text search across platform names
- ☐ "Export to CSV" button (gated — email required)
- ☐ Highlight enterprise chains with visual badge
- ☐ Link each row to corresponding POFOV coin/enterprise page
- ☐ SEO: target "blockchain comparison chart", "which blockchain supports smart contracts"
- ☐ Test: filter, sort, search all work; export triggers email gate

PF8 — Listmonk Email Sequences (3-4 hours)

- ☐ Create POFOV-specific welcome sequence in Listmonk (5 emails over 14 days):
  1. **Day 0:** Deliver gated content (report/comparison) + "Here's what else POFOV offers" (matrix, enterprise guides)
  2. **Day 3:** "3 things most teams get wrong choosing a blockchain platform" — educational, positions you as expert
  3. **Day 7:** "Private vs public blockchain: when each one wins" — links to amiheines.com blog post
  4. **Day 10:** Case study: "How ChainVault uses Antelope for wire fraud prevention" — links to amiheines.com/portfolio
  5. **Day 14:** "Want help evaluating blockchain for your project? → Book a free 30-min discovery call" → amiheines.com/contact
- ☐ Create weekly digest template:
  - New coins added to POFOV this week
  - Enterprise blockchain news highlights
  - Featured comparison of the week
  - Every 4th issue: consulting CTA
- ☐ Configure sending: welcome sequence auto-triggered on POFOV list signup
- ☐ Test: full sequence delivery, links work, unsubscribe works

PF9 — Category Deep Dive Pages (5-6 hours)

- ☐ Create `/category/[name]` pages for all 17 categories:
  - Category description and significance
  - Curated top 10 picks in the category (from POFOV data, ranked by market cap + editorial opinion)
  - Key trends in this category (editorial — 2-3 paragraphs)
  - "Enterprise applicability" assessment (high/medium/low)
  - Downloadable PDF summary (gated)
  - Grid of all coins in this category (linking to individual pages)
- ☐ Prioritize enterprise-relevant categories first:
  1. Enterprise, 2. Infrastructure, 3. Identity, 4. Data/Oracles, 5. Storage, 6. Payments
- ☐ SEO: "best enterprise blockchain 2026", "blockchain for identity management", etc.
- ☐ Link from main navigation and homepage category filter

PF10 — POFOV Monitoring (3-4 hours)

- ☐ Install `prom-client` in pofov.com Astro backend:
  - `pofov_page_views_total{page_type, slug}` — views by page type (coin, compare, matrix, enterprise)
  - `pofov_email_signups_total{source}` — signups by source (modal, inline, gated content)
  - `pofov_pdf_downloads_total{slug}` — PDF report downloads
  - `pofov_comparison_views_total{pair}` — which comparisons are popular
  - Common: `http_requests_total`, `http_request_duration_seconds`
- ☐ Add metrics to scraper FastAPI:
  - `scraper_job_duration_seconds{type}` — scrape job duration
  - `scraper_coins_processed_total{type}` — coins processed per scrape
  - `scraper_errors_total{type, error}` — scrape errors
- ☐ Expose `/metrics` on both
- ☐ Create POFOV Grafana dashboard:
  - Traffic: page views/day, top coins, top comparisons
  - Conversions: email signups/day, PDF downloads, signup sources
  - Scraper health: last job run, duration, errors, coins processed
- ☐ Add pofov.com + scrapecmc.pofov.com to Uptime Kuma

PF11 — Scraper Improvements (time permitting, 2-3 hours)

- ☐ Add cron scheduling to scraper (weekly auto-rescrape on Sunday 02:00):
  - Phase 1: listings refresh
  - Phase 2: deep scrape (new coins only, unless full rescrape flag)
  - Phase 3: AI questions for new coins
- ☐ Add new enterprise-focused AI questions:
  - "Is [coin] suitable for enterprise private deployments?"
  - "What are the storage capabilities of [coin] blockchain?"
- ☐ Add structured capability classification to AI output:
  - Parse AI answers to extract: has_smart_contracts, has_storage, has_privacy, enterprise_ready (boolean fields)
  - Store in new structured field alongside raw Q&A
  - Feed into capability matrix (PF5)

Outreach + Content (4 hours)

Outreach (2-3 hours)

- ☐ Send 10 new LinkedIn messages
- ☐ Pipeline management
- ☐ Post 3-5 LinkedIn insights
- ☐ Running total: 155-210 outreach messages
- ☐ Share POFOV matrix page on LinkedIn: "I mapped the capabilities of 50+ blockchain platforms"

Content (1-2 hours)

- ☐ Publish blog post #14
  - Candidate: "I Asked AI to Evaluate Every Major Blockchain for Enterprise Use — Here's the Ranking"
  - Candidate: "Building an AI-Powered Comparison Tool for Blockchain Platforms"
- ☐ Share on LinkedIn with link to POFOV

End-of-Week Checkpoints

- ☐ POFOV complete:
  - Email capture + gating + comparison tool + PDF reports + matrix + enterprise pages + category pages

  - Listmonk sequences configured and tested
  - Monitoring + Grafana dashboard live
  - Scraper auto-scheduling (if completed)

☐ **POFOV is now a full lead generation engine**
☐ 155-210 outreach messages total
☐ Track: POFOV signups this week, gated content unlocks, consulting CTAs clicked

Week 16 — ChainTalk: Smart Contract + Backend Core

Dates: _____ to _____ **Focus:** 80% ChainTalk new build / 20% outreach + content **Target hours:** 20

ChainTalk — Blockchain Comment System (16 hours)

CT1 — Smart Contract: chaintalk.core (5-6 hours)

Contract Code (3-4 hours)

- ☐ Create project structure: `chaintalk/blockchain/contracts/chaintalk.core/`
- ☐ Write `chaintalk.core.hpp` — tables:
  - `sites` (site_id, owner, domain, display_name, created_at, is_active, mod_mode, theme)
  - `comments` (comment_id, site_id, page_path, author, parent_id, body, body_hash, created_at, is_hidden)
  - `users` (user_id, account, display_name, email_hash, avatar_url, created_at)
  - `votes` (vote_id, comment_id, voter, value, created_at)
  - `moderations` (mod_id, comment_id, moderator, action, reason, created_at)
  - Indices: by_site_page (comments), by_author, by_comment (votes), by_moderator
- ☐ Write `chaintalk.core.cpp` — actions:
  - `regsite` (self auth — register domain)
  - `updatesite` (site owner — update settings)
  - `reguser` (self auth — register user account)
  - `postcomment` (user auth — post or reply, validate site_id + page_path)
  - `editcomment` (author auth — edit body, store new hash, history preserved via Hyperion)
  - `vote` (user auth — +1/-1, one vote per user per comment, toggle on re-vote)
  - `moderate` (site owner auth — hide/unhide/flag with reason)
- ☐ Write `CMakeLists.txt`

Compile + Deploy (1-2 hours)

- ☐ Compile with Antelope CDT
- ☐ Create `chaintalkco` account on Verarta blockchain
- ☐ Deploy WASM + ABI
- ☐ Test all actions via cleos:
  - Register a site
  - Register users
  - Post comments and replies (verify threading via parent_id)
  - Vote on comments
  - Edit a comment (verify old version accessible via Hyperion history)
  - Moderate: hide, unhide, verify moderation record
- ☐ Verify all actions and deltas in Hyperion

CT2 — Backend Foundation (4-5 hours)

- ☐ Create `chaintalk/backend/` Astro 5 SSR project
- ☐ Install dependencies: @astrojs/node, pg, redis, jsonwebtoken, bcrypt, prom-client, zod
- ☐ Create PostgreSQL database `chaintalk`
- ☐ Write migration:

```
sites (id, domain, owner_account, api_key_hash, display_name, settings, created_at)
users (id, account, email, password_hash, display_name, avatar_url, created_at)
sessions (id, user_id, token_hash, expires_at, created_at)
comments_cache (id, blockchain_comment_id, site_id, page_path, author_account, author_name,
```

- ☐ Create core libs:
  - `db.ts` — PostgreSQL pool
  - `redis.ts` — Redis client

  - `auth.ts` — JWT + bcrypt + sessions
  - `antelope.ts` — chain client (Verarta cluster)
  - `api-key.ts` — API key generation + validation for site owners

- ☐ Create auth middleware (JWT for users, API key for site embed requests)
- ☐ Astro config: SSR, Node adapter, port 4323
- ☐ PM2 ecosystem config
- ☐ Test: starts, connects to DB + Redis + chain

CT3 — Comment CRUD API (5-6 hours)

- ☐ `GET /api/comments/:siteId/:pageId` — get comments for a page:
  - Paginated (default 50 per page)
  - Returns threaded tree structure (comments with nested replies)
  - Include: author display name, avatar, vote counts, created_at, blockchain tx link
  - Include moderation status (hidden comments show "[hidden by moderator — reason]")
  - Cache in Redis (invalidate on new comment/vote/moderation)
- ☐ `POST /api/comments` — post a comment:
  - Validate: site_id exists, page_path non-empty, body 1-2000 chars
  - If parent_id > 0: validate parent exists and belongs to same site+page
  - Push `postcomment` action to chain
  - Insert into comments_cache
  - Invalidate Redis cache for this page
  - Return: new comment with blockchain tx ID
- ☐ `PUT /api/comments/:id` — edit a comment:
  - Validate: user is author
  - Push `editcomment` action to chain (new body + hash)
  - Update comments_cache
  - Invalidate cache
- ☐ `POST /api/comments/:id/vote` — vote:
  - Validate: user is not author (can't self-vote)
  - Push `vote` action to chain
  - Update vote counts in cache
  - Return: new vote counts
- ☐ `POST /api/comments/:id/moderate` — moderate:
  - Validate: user is site owner
  - Push `moderate` action to chain
  - Update comments_cache (is_hidden)
  - Return: moderation record
- ☐ Test: full CRUD cycle — post, reply, edit, vote, moderate — all reflected on chain + cache

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Outreach + Content (4 hours)

Outreach (3 hours)

- ☐ Send 10 new LinkedIn messages
- ☐ Pipeline management, close deals
- ☐ Post 3-5 LinkedIn insights
- ☐ Running total: 165-220 outreach messages

Content (1 hour)

- ☐ Publish blog post #15
  - Candidate: "Writing a Smart Contract for a Comment System — Why Blockchain Comments Can't Be Silently Deleted"
- ☐ Share on LinkedIn

End-of-Week Checkpoints

- ☐ chaintalk.core contract deployed and tested on Verarta chain
- ☐ ChainTalk backend running with full comment CRUD API
- ☐ Comments stored on chain + cached in PostgreSQL + cached in Redis
- ☐ Threading, voting, and moderation all working
- ☐ 165-220 outreach messages total

Week 17 — ChainTalk: Site Registration + User Auth + Embed Widget Start

Dates: _____ to _____ **Focus:** 80% ChainTalk / 20% outreach + content **Target hours:** 20

ChainTalk Expansion (16 hours)

CT4 — Site Registration + Domain Verification (3-4 hours)

- ☐ `POST /api/sites/register` — register a new site:
  - Input: domain, display_name, email
  - Generate API key (random 32-byte hex)
  - Generate verification token (random string)
  - Store in sites table (unverified state)
  - Return: API key + verification instructions (DNS TXT or meta tag)
- ☐ `GET /api/sites/:id/verify` — verify domain ownership:
  - **DNS TXT method:** check for TXT record `chaintalk-verify=<token>` on domain
  - **Meta tag method:** fetch domain homepage, check for `<meta name="chaintalk-verify" content="<token>">`
  - On verification: push `regsite` action to chain, update DB, mark verified
- ☐ `GET /api/sites/:id` — get site details (settings, stats, verification status)
- ☐ `PUT /api/sites/:id` — update site settings (moderation mode, theme, display name)
- ☐ API key middleware: validate API key header on embed requests
- ☐ Test: register site → verify via both methods → site active on chain

CT5 — User Auth (3-4 hours)

- ☐ `POST /api/auth/register` — email + password + display name:
  - Create blockchain account for user (auto-generated name)
  - Push `reguser` action to chain
  - Hash password, store in DB
  - Send verification email
  - Return: session JWT
- ☐ `POST /api/auth/login` — email + password:
  - Validate credentials
  - Create session
  - Return: JWT
- ☐ `GET /api/auth/session` — get current user info
- ☐ `POST /api/auth/logout` — terminate session
- ☐ Anonymous posting mode:
  - If site settings allow anonymous: accept `display_name` without registration
  - Create ephemeral blockchain account per anonymous commenter
  - Anonymous comments marked as such in UI
- ☐ JWT token format: includes user_id, account name, display_name
- ☐ Test: register, verify email, login, post comment, logout

CT6 — Embed Widget JS Loader (4-5 hours)

- ☐ Create `embed.js` — vanilla JavaScript, no framework:
  - Tiny loader (<5KB minified): reads `data-*` attributes from host `<div>`
  - Creates iframe pointing to `chaintalk.amiheines.com/widget/:siteId/:pageId`
  - Auto-resize iframe height based on content (via postMessage)
  - Pass theme (light/dark) and accent color to iframe
  - Handle error states: invalid site, API down, rate limited
- ☐ Serve from `GET /api/embed.js` with long cache headers + versioning
- ☐ PostMessage protocol between iframe and host:

  - `chaintalk:resize` — iframe sends height, host resizes
  - `chaintalk:ready` — iframe loaded
  - `chaintalk:error` — error state
  - `chaintalk:auth` — user logged in/out

- ☐ Test embed on a local HTML page:

```
<script src="http://localhost:4323/api/embed.js" async></script>
<div id="chaintalk" data-site-id="test" data-page-id="/test" data-theme="light"></div>
```

- ☐ Verify: iframe loads, resizes correctly, theme applies

CT7 — Embed Widget Iframe UI — Start (4-5 hours)

Build the iframe-hosted widget UI (this continues into week 18):

Comment List (3-4 hours)

- ☐ Create widget page: `/widget/:siteId/:pageId`
  - Minimal, self-contained HTML/CSS/JS (no heavy framework in iframe)
  - Or: lightweight React/Preact build specifically for widget
- ☐ Render comment list:
  - Threaded display (nested replies with indentation, max 3 levels)
  - Each comment shows: avatar, display name, time ago, body text, vote buttons (+/-), vote count
  - Blockchain receipt icon ([chain]) linking to transaction on Verarta explorer
  - Hidden comments: show "[Comment hidden by moderator — reason: ...]" with link to moderation tx
  - "Load more" pagination at bottom
- ☐ Sort options: newest first, oldest first, most voted

Auth State in Widget (1 hour)

- ☐ Login/register form inline within widget:
  - "Sign in to comment" link → expand inline form (email + password + display name)
  - Or "Comment as guest" if site allows anonymous
  - JWT stored in iframe localStorage (scoped to chaintalk.amiheines.com)
- ☐ Show logged-in state: "Commenting as [name]" with logout link

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Outreach + Content (4 hours)

Outreach (3 hours)

- ☐ Send 10 new LinkedIn messages
- ☐ Pipeline management
- ☐ Post 3-5 LinkedIn insights
- ☐ Running total: 175-230 outreach messages

Content (1 hour)

- ☐ Publish blog post #16
  - Candidate: "Building an Embeddable Widget with Vanilla JS and iframes — No React Required"
  - Candidate: "I Built a Blockchain Alternative to Disqus — Here's Why"
- ☐ Share on LinkedIn

End-of-Week Checkpoints

- ☐ Site registration + domain verification working (DNS TXT + meta tag)
- ☐ User registration + login working
- ☐ Embed.js loader working — iframe loads on any HTML page

- ☐ Comment list rendering in widget iframe (threaded, with blockchain links)
- ☐ Auth inline in widget
- ☐ 175-230 outreach messages total

Week 18 — ChainTalk: Widget UI Completion + Landing Page

Dates: _____ to _____ **Focus:** 80% ChainTalk / 20% outreach + content **Target hours:** 20

ChainTalk Widget + Landing (16 hours)

CT7 continued — Embed Widget Iframe UI Completion (4-5 hours)

Comment Posting (2-3 hours)

- ☐ Post comment form:
  - Textarea with character counter (max 2000)
  - "Post to blockchain" button with loading state
  - After post: new comment appears immediately (optimistic UI), blockchain tx link appears when confirmed
  - Reply: "Reply" button on each comment → expand reply form inline below comment
- ☐ Voting UI:
  - Up/down arrow buttons per comment
  - Current vote count between arrows
  - Highlight user's own vote
  - Prevent self-voting
  - Loading state during blockchain transaction
- ☐ Error handling:
  - Network errors: "Failed to post. Retry?"
  - Rate limiting: "Please wait before posting again"
  - Auth required: "Sign in to comment"

Theming + Polish (2 hours)

- ☐ Light theme (default): white background, dark text
- ☐ Dark theme: dark background, light text
- ☐ Customizable accent color via `data-accent` attribute
- ☐ CSS variables for all theme tokens (easy for Pro tier customization later)
- ☐ Responsive: works on mobile (320px) to desktop
- ☐ Loading skeleton while comments load
- ☐ Empty state: "Be the first to comment!"
- ☐ "Powered by ChainTalk [chain]" footer link (links to landing page)

CT8 — ChainTalk Landing Page (4-5 hours)

- ☐ Create marketing page at `chaintalk.amiheines.com`:
  - **Hero:** "Comments that can't be silently deleted." — subhead: "Drop-in blockchain-backed comments for any website."
  - **How it works:** 3-step: 1. Add script tag, 2. Comments stored on blockchain, 3. Every message has a receipt
  - **Live demo:** Embedded ChainTalk widget right on the landing page (meta!)
  - **Comparison table:** ChainTalk vs Disqus vs Commento (the pitch from section 11.10.5)
  - **Pricing:** Free / Pro / Enterprise / Self-hosted tiers
  - **Setup wizard:** "Get started in 2 minutes" → register site → get embed code → copy-paste
  - **FAQ:** How does it work? Is it really immutable? What about spam? Can I moderate? How fast?
  - **Footer:** "Built by Ami Heines — Blockchain Infrastructure Consultant" → amiheines.com
- ☐ SEO: title, meta description, Open Graph
- ☐ Matomo tracking
- ☐ Test: page loads, demo widget works, setup wizard generates valid embed code

CT9 — Admin Dashboard (5-6 hours)

- ☐ Site owner dashboard at `chaintalk.amiheines.com/dashboard`:
  - **Stats:** total comments, comments today, active users, vote count

- **Comment moderation queue:**
  - Flagged comments requiring review
  - Recent comments (newest first)
  - Hide/unhide actions with reason input
  - Bulk moderation (select multiple → hide)
- **Site settings:**
  - Display name, moderation mode (pre-approve / post-moderate / open)
  - Anonymous commenting toggle
  - Theme selection (light/dark/auto)
  - Accent color picker
- **API key management:**
  - Show current API key (masked, reveal on click)
  - Regenerate API key (with confirmation warning)
- **Embed code generator:**
  - Preview embed code with current settings
  - Copy-to-clipboard button
  - Live preview of widget with current theme
- **Pages list:** all pages with comments, sorted by recent activity

☐ Dashboard auth: site owner login (same auth system, site_owner flag)
☐ Test: full dashboard flow — login, view stats, moderate comment, update settings, copy embed code

---

Outreach + Content (4 hours)

Outreach (3 hours)

☐ Send 10 new LinkedIn messages
☐ Pipeline management, close deals
☐ Post 3-5 LinkedIn insights
☐ Running total: 185-240 outreach messages
☐ **Revenue milestone check:** Should be at $2K+/month by now from outreach + freelance + any POFOV leads

Content (1 hour)

☐ Publish blog post #17 or YouTube video
  - Candidate: "I Built a Blockchain Alternative to Disqus — Here's Why and How" (great video)
  - Candidate: "Why Your Comment Section Should Be on a Blockchain (5 Minutes for CTOs)"
☐ Share on LinkedIn + Product Hunt consideration for ChainTalk launch

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End-of-Week Checkpoints

☐ ChainTalk widget fully functional: post, reply, vote, moderate, themed
☐ Landing page live at chaintalk.amiheines.com with live demo
☐ Admin dashboard working: stats, moderation, settings, embed code generator
☐ Setup wizard: register → verify → get embed code → works on any site
☐ 185-240 outreach messages total

Week 19 — ChainTalk: Monitoring + Deploy + Integrate into POFOV

Dates: _____ to _____ **Focus:** 60% ChainTalk deploy + monitoring / 30% integrations / 10% outreach **Target hours:** 20

ChainTalk Deploy + Monitoring (12 hours)

CT10 — Prometheus Metrics (2-3 hours)

- ☐ Install `prom-client` in ChainTalk backend:
  - `chaintalk_comments_posted_total{site}` — comments per site
  - `chaintalk_votes_total{site, direction}` — votes by direction
  - `chaintalk_moderations_total{site, action}` — moderation actions
  - `chaintalk_sites_registered_total` — total registered sites
  - `chaintalk_active_users_total` — users who posted in last 24h
  - `chaintalk_embed_loads_total{site}` — widget loads (from embed.js requests)
  - `blockchain_transaction_duration_seconds` — chain tx latency
  - Common: `http_requests_total`, `http_request_duration_seconds`, `app_errors_total`
- ☐ Expose `/metrics` endpoint
- ☐ Add to Prometheus scrape config
- ☐ Test: metrics visible in Prometheus

CT11 — Grafana Dashboard (2-3 hours)

- ☐ Create ChainTalk dashboard:
  - **Overview:** total sites, total comments, total users, comments today
  - **Activity:** comments posted/day, votes/day, new users/day
  - **Sites:** top sites by comment volume, embed loads/day
  - **Moderation:** moderation actions/day, flagged comments pending
  - **API:** request rate, latency p50/p95/p99, error rate
  - **Blockchain:** tx latency, tx volume, chain health
- ☐ Provision as JSON dashboard file
- ☐ Test: panels render with data from test comments

CT12 — Deployment (2-3 hours)

- ☐ Create nginx config for chaintalk.amiheines.com:
  - Landing page + dashboard: proxy to port 4323
  - API: proxy to /api/
  - Embed script: proxy to /api/embed.js with aggressive cache headers
  - Widget iframe: proxy to /widget/
  - SSL via certbot
  - CORS headers for embed cross-origin requests
- ☐ PM2 ecosystem config for chaintalk-backend
- ☐ Create deploy script
- ☐ Production .env from template
- ☐ DNS: A record for chaintalk.amiheines.com
- ☐ Deploy to server
- ☐ Add to Uptime Kuma
- ☐ Test: full flow from external site embedding to comment posting

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Integrations (6 hours)

CT13 — Integrate ChainTalk into POFOV (3-4 hours)

- ☐ Register pofov.com as a site in ChainTalk
- ☐ Verify domain ownership

- ☐ Add ChainTalk embed to every POFOV coin page template:
  - Place below coin content, above footer
  - `data-page-id` = coin slug (e.g., `/project/bitcoin`)
  - Light theme to match POFOV design
- ☐ Add ChainTalk embed to comparison pages
- ☐ Add ChainTalk embed to enterprise pages
- ☐ Seed 5-10 starter comments on popular coin pages (from your account — kickstart discussion)
- ☐ Test: visit coin page → see comment widget → post a comment → appears on chain
- ☐ Matomo: track ChainTalk engagement on POFOV pages

CT14 — Integrate ChainTalk into amiheines.com (1-2 hours)

- ☐ Register amiheines.com as a site in ChainTalk
- ☐ Add ChainTalk embed to blog post template
- ☐ `data-page-id` = blog post slug
- ☐ Seed 1-2 comments on existing blog posts
- ☐ Test: blog post → comments visible → can post
- ☐ Verify: "Powered by ChainTalk" links to chaintalk.amiheines.com

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Outreach + Content (2 hours)

Outreach (1-2 hours)

- ☐ Send 5-10 LinkedIn messages
- ☐ Pipeline management
- ☐ Post 2-3 LinkedIn insights
- ☐ Running total: 190-250 outreach messages

Content (1 hour)

- ☐ Publish blog post #18
  - Candidate: "Adding Blockchain Comments to 1,000 Pages in One Deploy" (CT13 experience)
  - Candidate: "Why I Added Blockchain Comments to 1,000 Crypto Research Pages (And What Happened)"
- ☐ Share on LinkedIn

End-of-Week Checkpoints

- ☐ ChainTalk fully deployed at chaintalk.amiheines.com
- ☐ Monitoring: metrics, Grafana dashboard, Uptime Kuma
- ☐ ChainTalk embedded on all POFOV pages (1,000+ pages with blockchain comments)
- ☐ ChainTalk embedded on amiheines.com blog posts
- ☐ Both sites verified and active in ChainTalk
- ☐ 190-250 outreach messages total

Week 20 — ChainTalk Demo Site + Final Integration + Portfolio Update

Dates: _____ to _____ **Focus:** 40% ChainTalk demo / 40% portfolio + cross-project / 20% outreach + content
Target hours: 20

ChainTalk Demo + Polish (8 hours)

CT15 — Demo Site (3-4 hours)

- ☐ Create demo blog at `demo.chaintalk.amiheines.com`:
  - Simple Astro site with 5 blog posts about blockchain/tech topics
  - Each post has ChainTalk widget embedded
  - Pre-seeded with demo comments showing:
    - Threaded replies (2-3 levels deep)
    - Upvoted/downvoted comments
    - At least 1 moderated (hidden) comment with visible moderation reason
    - Mix of registered and anonymous (if enabled) commenters
  - Header: "This is a demo of ChainTalk — blockchain-backed comments. Try it yourself!"
  - Each comment has visible blockchain receipt link
- ☐ Nginx config + certbot for demo.chaintalk.amiheines.com
- ☐ Link from ChainTalk landing page: "See it in action → demo.chaintalk.amiheines.com"
- ☐ Test: visitors can post real comments on demo site

ChainTalk Polish (2-3 hours)

- ☐ Review and fix any bugs found during POFOV/amiheines integration
- ☐ Performance: ensure embed.js loads in <100ms, widget renders in <500ms
- ☐ Verify CORS works from all integrated domains
- ☐ Rate limiting on comment posting (5 comments/min per user)
- ☐ Spam protection: basic keyword filter + honeypot field in post form
- ☐ Accessibility: keyboard navigation, ARIA labels, screen reader support

PF12 — POFOV ChainTalk Review (2 hours)

- ☐ Review ChainTalk integration on POFOV:
  - Test on 5-10 different coin pages
  - Verify loading speed, theme consistency
  - Check mobile rendering
  - Verify page_id uniqueness across all pages
- ☐ Monitor first organic comments on POFOV (if any)
- ☐ Seed additional starter comments/questions on high-traffic coin pages
- ☐ Add "Join the discussion below" CTA text above ChainTalk widget on POFOV pages

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Cross-Project Integration + Portfolio (8 hours)

Portfolio Page Update (3-4 hours)

- ☐ Update amiheines.com/portfolio with all 5 projects:
  - **ChainVault:** Screenshot, description, demo link, Grafana dashboard screenshot
  - **Verarta:** Screenshot, gallery link, demo link, Grafana dashboard screenshot
  - **Parking Tickets:** Screenshot, description, demo link, Grafana dashboard screenshot
  - **POFOV:** Screenshot, link, description of scale (1,000+ pages)
  - **ChainTalk:** Screenshot, demo link, embed code example, landing page link
- ☐ Each project shows: tech stack badges, "View Grafana Dashboard" link (if public), "Try Demo" button
- ☐ Add "Monitoring" section: screenshot of Grafana overview + link to status.amiheines.com

- ☐ Add “Open-Source Stack” section: cost savings table from plan section 11.8
- ☐ SEO: update meta tags, Open Graph images for portfolio page

Status Page Review (1 hour)

- ☐ Verify Uptime Kuma monitors all 5 apps + infrastructure:
  - amiheines.com
  - chainvault.amiheines.com
  - verarta.com
  - parking.amiheines.com
  - pofov.com
  - chaintalk.amiheines.com
  - monitoring.amiheines.com
  - lists.amiheines.com
  - matomo.amiheines.com
- ☐ Verify status.amiheines.com is publicly accessible and shows all monitors
- ☐ Check uptime percentages — fix any intermittent issues

Cross-Promotion Audit (2 hours)

- ☐ Verify all cross-links work:
  - amiheines.com → all project demos
  - POFOV → amiheines.com consulting CTAs
  - ChainTalk → amiheines.com footer link
  - All projects → status.amiheines.com
  - Blog posts → relevant project links
- ☐ Verify all Matomo tracking fires on all sites
- ☐ Review Listmonk subscriber growth: amiheines.com list + POFOV list
- ☐ Check email sequence delivery rates and open rates
- ☐ Fix any broken links or missing tracking

Services Page Update (1 hour)

- ☐ Update amiheines.com/services with ChainTalk self-hosted tier:
  - Add: “$2,500 — ChainTalk Self-Hosted Setup” (deploy blockchain comment system on your infrastructure)
  - Add: “$99/month — ChainTalk Enterprise” managed tier
- ☐ Add ChainTalk demo to discovery call script as a live demo option
- ☐ Update proposal template with ChainTalk as a portfolio reference

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Outreach + Content (4 hours)

Outreach (3 hours)

- ☐ Send 10 new LinkedIn messages
- ☐ Pipeline management
- ☐ Post 3-5 LinkedIn insights
- ☐ Running total: 200-260 outreach messages
- ☐ **Share portfolio update on LinkedIn:** “I just finished building 5 blockchain apps — all open source, all monitored, all on one server”

Content (1 hour)

- ☐ Publish blog post #19 — milestone post:
  - “How I Built 5 Blockchain Apps in 20 Weeks Using Claude Code”
  - Or: “My Full Stack: 5 Blockchain Apps, Open-Source Monitoring, One Server, $0 in Licensing”
- ☐ Share on LinkedIn, cross-post key points to POFOV blog

End-of-Week Checkpoints

- ☐ ChainTalk demo site live with seeded conversations
- ☐ All 5 projects polished and demo-ready
- ☐ amiheines.com/portfolio fully updated with all projects
- ☐ All cross-links verified
- ☐ status.amiheines.com monitoring all services
- ☐ Matomo tracking on all sites
- ☐ 200-260 outreach messages total
- ☐ **Milestone blog post published**

Week 21 — Final Integration, Review, and Acceleration Plan

Dates: _____ to _____ **Focus:** 30% final polish / 30% revenue acceleration / 40% content + outreach scaling
Target hours: 20

Final Polish + Review (6 hours)

Full System Smoke Test (2 hours)

- ☐ Walk through every project as a first-time visitor:
  - **amiheines.com:** Landing → Services → Portfolio → Blog → Contact → Newsletter signup
  - **ChainVault demo:** Click demo → explore deals → documents → audit trail → wire verification
  - **Verarta demo:** Click demo → explore artworks → gallery → provenance → blockchain explorer
  - **Parking demo:** Admin login → dashboard → tickets → employee view → public payment page
  - **POFOV:** Browse coins → enterprise page → comparison tool → download PDF → email captured
  - **ChainTalk:** Landing page → try demo → post comment → see blockchain receipt → admin dashboard
- ☐ Fix any UX issues, broken links, slow pages, or confusing flows
- ☐ Mobile test: repeat above on phone (or browser dev tools mobile mode)

Grafana Dashboard Review (1 hour)

- ☐ Review all Grafana dashboards with real data:
  - Server, Docker, Blockchain, ChainVault, Verarta, Parking, POFOV, ChainTalk, PostgreSQL, Redis, ES, Nginx
- ☐ Fix any panels with no data or errors
- ☐ Create a "Portfolio Overview" dashboard:
  - Combined view: all 5 apps on one page
  - Total blockchain transactions across all apps
  - Combined API request rate and error rate
  - Status badges for all services
- ☐ Take screenshots of key dashboards for portfolio page and blog posts

Documentation Pass (1 hour)

- ☐ Update each project's README with current state
- ☐ Ensure DEPLOYMENT.md files are current for all projects
- ☐ Update CLAUDE.md files with any new architecture decisions
- ☐ Clean up any leftover TODO_AMIH.md items

Security Audit (2 hours)

- ☐ Check all production .env files: no secrets in git, correct permissions
- ☐ Verify all API endpoints require proper auth
- ☐ Check rate limiting on all login/payment/submission endpoints
- ☐ Verify Stripe webhook signature validation
- ☐ Check CORS configs: only allow expected origins
- ☐ Verify nginx: no open proxies, rate limiting active, SSL A+ rating
- ☐ Check ChainTalk embed: no XSS vectors, iframe sandbox attributes
- ☐ Run `npm audit` on all projects, fix any critical vulnerabilities

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Revenue Acceleration (6 hours)

Revenue Review + Strategy Adjustment (2 hours)

- ☐ Compile revenue metrics for weeks 1-21:
  - Total outreach messages sent

    - Response rate
    - Discovery calls conducted
    - Proposals sent
    - Deals closed (number + total value)
    - Freelance platform revenue
    - POFOV leads generated
    - ChainTalk signups
- ☐ Identify what's working → double down
- ☐ Identify what's not working → drop or adjust
- ☐ Set revenue targets for weeks 22-30

Testimonial Collection (1 hour)

- ☐ Request testimonials from any customers served so far
- ☐ Request LinkedIn recommendations from satisfied clients
- ☐ Ask: "Can I use your company as a case study on my portfolio?"
- ☐ Add testimonials to amiheines.com landing page and /services

Referral System (1 hour)

- ☐ Reach out to satisfied customers: "Do you know anyone else who could benefit from tamper-proof records?"
- ☐ Offer referral incentive: 10% off their next engagement for each referral that converts
- ☐ Add to Listmonk: occasional referral request in newsletter

Case Study Creation (2 hours)

- ☐ Write 1-2 detailed case studies from real customer engagements (if any)
- ☐ Or: write detailed case studies of your own projects (ChainVault, Parking Tickets) framed as client work:
    - Problem → Solution → Architecture → Results → Monitoring
    - Include Grafana screenshots as proof of production quality
- ☐ Publish on amiheines.com/portfolio and as blog posts
- ☐ Format as PDF for use in proposals

Content + Outreach Scaling (8 hours)

Content Batch Production (4-5 hours)

- ☐ Record 2-3 YouTube videos (batch recording session):
    - Video 1: "How I Built 5 Blockchain Apps in 20 Weeks Using Claude Code" (portfolio tour)
    - Video 2: "Enterprise Monitoring for $0: Grafana + Prometheus on a Single Server" (screen share)
    - Video 3: "I Built a Blockchain Alternative to Disqus — Here's Why" (ChainTalk walkthrough)
- ☐ Edit and publish at least video 1 this week
- ☐ Create blog post versions of each video
- ☐ Schedule: 1 video/week for next 3 weeks (batch gives buffer)

Outreach at Scale (3 hours)

- ☐ Send 15-20 LinkedIn messages (bigger batch — now have stronger portfolio to reference)
- ☐ Include portfolio links and demo links in messages
- ☐ Post LinkedIn insight referencing video/case study
- ☐ Comment on 15+ prospect posts
- ☐ Running total: 215-280 outreach messages
- ☐ Consider: LinkedIn premium or Sales Navigator for better targeting

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Ongoing Operations Checklist

From this week forward, establish a weekly rhythm:

Weekly Maintenance (2-3 hours/week)

- ☐ Check Grafana dashboards for anomalies
- ☐ Review Alertmanager — any alerts this week?
- ☐ Check Uptime Kuma — any downtime events?
- ☐ Review Matomo analytics across all sites
- ☐ Check Listmonk subscriber growth + email open rates
- ☐ Check POFOV scraper health (if auto-schedule running)
- ☐ Review ChainTalk: new sites registered? Comment volume?
- ☐ Clear Docker logs/old images if disk is growing

Weekly Growth (10+ hours/week)

- ☐ 10-15 LinkedIn outreach messages
- ☐ 3-5 LinkedIn posts/insights
- ☐ 1 blog post published
- ☐ 1 YouTube video published (from batch)
- ☐ 1 newsletter sent via Listmonk
- ☐ Follow up on pipeline, close deals

End-of-Week 21 Checkpoints — THE BIG PICTURE

What's Live

- ☐ [OK] amiheines.com — main hub, blog, portfolio, services, newsletter
- ☐ [OK] chainvault.amiheines.com — wire fraud prevention, demo mode
- ☐ [OK] verarta.com — art provenance, public gallery, demo mode
- ☐ [OK] parking.amiheines.com — municipal parking tickets, full flow
- ☐ [OK] pofov.com — 1,000+ page SEO engine, email capture, comparisons
- ☐ [OK] chaintalk.amiheines.com — embeddable blockchain comments, demo site
- ☐ [OK] monitoring.amiheines.com — Grafana dashboards for everything
- ☐ [OK] status.amiheines.com — public uptime status page
- ☐ [OK] lists.amiheines.com — Listmonk email marketing
- ☐ [OK] matomo.amiheines.com — privacy-first analytics

What's Monitored

- ☐ [OK] Every app has Prometheus metrics + Grafana dashboard
- ☐ [OK] Structured logging → Loki for all apps
- ☐ [OK] Alert rules → email notifications via Alertmanager
- ☐ [OK] Uptime monitoring → public status page

Revenue Pipeline

- ☐ 215-280 total outreach messages sent over 21 weeks
- ☐ Blog: 19+ posts published
- ☐ YouTube: 3+ videos published (more in pipeline)
- ☐ Listmonk: subscribers growing from amiheines.com + POFOV
- ☐ POFOV: generating organic search traffic + email leads
- ☐ ChainTalk: sites registering, proof of concept for embeddable blockchain
- ☐ Target: $2K+/month recurring from consulting + freelance + managed services

What Comes Next (Weeks 22+)

- Shift time allocation: 30% maintenance + dev, 70% outreach + content + sales
- Scale content: 2 videos/week now that portfolio is complete
- Pursue managed infrastructure contracts ($2-5K/month recurring)
- ChainTalk Product Hunt launch
- POFOV weekly auto-rescrape + newsletter automation
- Upsell existing customers to higher tiers
- Consider: conference talks, podcast appearances, guest blog posts for link building